Software Requirements Specification (SRS) for the LavaRápido System

Web and Mobile Booking and Management Platform for a Car Wash

Juan Camilo Guilombo Osorio, Brayan Estiven Patiño Cabrera and Merari Urbano Quintero

Centro de la Industria, la Empresa y los Servicios, Servicio Nacional de Aprendizaje (SENA), Huila Regional

Software Analysis and Development Technologist program (ADSO), group (ficha) 3145556

Instructor: Karol Daniela Correa

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Version Control

Table 1 *Document Version History*

Version	Date	Author	Description of the change
V2	2026-09-08	Development team	First version of the SRS. It described a mobile service with coverage zones and an online payment gateway.
V3	2026-10-06	Development team	Actual scope: a single fixed car wash, payment by QR code or bank transfer with a receipt, microservice architecture, requirements RF-001 to RF-026 with their status, and new use cases.
V4	2026-10-08	Development team	Restructured according to IEEE 830 and ISO/IEC/IEEE 29148, in APA 7 format. Status checked against the code as of October 8, 2026. New requirements RF-027 to RF-035 and use cases CU26 to CU29, business rules, flows and life cycles, data requirements, classified constraints, verifiable acceptance criteria, traceability matrix and APA references.
V5	2026-10-09	Development team	Verification with the system running (six services, gateway and web), and a fix to a gateway route and to a styling error in promotions. Use case and state diagrams in Mermaid, web screenshots in every module and an updated count of automated tests.
V6	2026-10-15	Development team	Status checked against the code as of October 15, 2026. Account lockout (RF-004b), automatic assignment (RF-008), operator choice by the customer (RF-010), payment history filters (RF-015) and operator availability (RF-022) are now implemented. New requirements RF-036 to RF-038 (inspection report, business data and operator profile) and use case CU30. Service start window, alert for bookings without an operator, notifications in each account's language, profile address, 57 tables, 601 automated tests and measurements for RNF-001 and RNF-008.
V7	2026-10-16	Development team	Complete review with the system running in Docker. The date and time the customer accepts the Terms and the Data Processing Policy are stored on the server (RR-01); HTTPS active on the public server; reason why a time is rejected (closed day, outside opening hours, time passed, or no free bay); 639 automated tests and 19 end-to-end tests passed.
V7.1	2026-10-17	Development team	Status verified against the code as of October 17, 2026: the inspection report (RF-036) is also managed from the mobile app, and the RNF-008 width check was extended to 60 web screens and 26 mobile app screens, with no overflow.
V7.2	2026-10-19	Development team	Continuous integration with GitHub Actions in the 10 repositories: every change is built and its tests are run. Continuous integration with Jenkins is removed from the pending items.
V7.3	2026-10-19	Development team	The administrator can change the role of an account (RF-028, CU27); an administrator account cannot be deactivated and an administrator cannot close their own account (RF-027).
V7.4	2026-10-19	Development team	New use cases CU31 (manage own account, RF-027) and CU32 (automatic processes: first administrator, notifications per event, reminders, alert of a booking without an operator and new version notice). There are now 32 use cases.
V7.5	2026-10-19	Development team	Tests of October 19, 2026 (655 automated, 19 end-to-end, and 42 functional cases, Table 58) and system screenshots taken again that day.

Version	Date	Author	Description of the change
V7.6	2026-10-20	Development team	The role change of an account is removed, because the team reverted it in the code (RF-028, CU27): the role is set when the account is created and the system does not offer to change it. An administrator account still cannot be deactivated and an administrator cannot close their own account. The domain events table (only events with a consumer travel through RabbitMQ), the notification types of RF-018 and the detail with a "Go to…" button of CU17, the service execution state diagram (Figure 6: no pause or issue, which are not used) and constraint RT-05 (public server on Azure) are corrected. RF-007 and CU07: the customer only cancels, without writing a reason, and rescheduling is done by the administrator (RF-033, CU29).

Abstract

This document specifies the requirements of LavaRápido, a web and mobile platform that organizes the operation of a single-site car wash: customer and vehicle registration, booking of time slots with opening-hours and bay validation, automatic or manual operator assignment, service execution and rating, vehicle inspection report, payment by QR code or bank transfer with receipt review, promotions, loyalty points and notifications. The specification follows the structure of IEEE 830 (IEEE, 1998) and ISO/IEC/IEEE 29148 (ISO/IEC/IEEE, 2018), and is based on the project's technical documentation (`vehicle-w-docs` repository) and on the code of its seven server components, reviewed on October 20, 2026. It defines 41 functional requirements with verifiable acceptance criteria, 12 non-functional requirements with their verification metric, 32 use cases, the business rules, the life cycles of the booking, the payment and the service execution, the data requirements, and the technical, data, operational and regulatory constraints. As of October 20, 2026, 40 functional requirements are implemented and 1 is partially implemented.

Keywords: requirements specification, car wash, bookings, microservices, use cases.

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1. Introduction

Many car washes still run on paper. Bookings come in by phone or customers show up without notice, time slots overlap, operator assignment depends on who happens to be available that day, and cash payments are not linked to any booking. LavaRápido brings these processes together: the customer registers their vehicles, books a service at a free time, pays by QR code or bank transfer and rates the service; the operator checks their assigned services and records their progress; and the administrator sets up the catalog, the opening hours and the bays, manages the operators and reviews payments.

1.1 Purpose

This document defines, completely and verifiably, what the LavaRápido system must do and under which quality conditions. It serves as an agreement between the development team, the SENA technical instructors and the person who manages the car wash, and as the basis for design, testing and product acceptance. Its structure follows the IEEE 830 recommended practice (IEEE, 1998) and the software requirements specification contents of ISO/IEC/IEEE 29148 (ISO/IEC/IEEE, 2018); the document format follows the publication manual of the American Psychological Association [APA] (2020).

1.2 Product Scope

The system consists of a web application, an Android mobile application and a REST API organized into six microservices behind an API Gateway. It runs a single car wash and covers the following functions:

- Registration, sign-in, password recovery and change, and account administration for three fixed roles: customer, operator and administrator.
- Registration of the customer's vehicles; the vehicle type determines the price of each service.
- Service catalog with price and duration by vehicle type, weekly opening hours with exceptions, and wash bays.
- Booking of time slots with availability validation, rescheduling and cancellation before the service starts.
- Automatic operator assignment when the booking is confirmed, optional operator choice by the customer and manual assignment by the administrator, with validation of shifts, absences and schedule overlaps.
- Start and finish of the service by the operator, and rating of the service by the customer.
- Vehicle inspection report for long services, shared with the customer through a link.
- Payment by QR code or bank transfer with a receipt reviewed by the administrator, and cash payment recorded by the administrator.
- Promotions with a percentage discount and a loyalty points program.
- Notifications in an inbox, by push and by email; a dashboard per role; visual theme and language per account; operational reports.

Out of scope for this version: the iOS application, home service, electronic invoicing with the Colombian tax authority (DIAN), supplies inventory control, online payment gateways, and a pre-production environment. The system is published on a single server (an Azure virtual machine with Docker Compose) for the academic review, not for the real operation of a car wash.

1.3 Definitions, Acronyms and Abbreviations

Table 2 *Domain Terms*

Term	Definition
User	Person with an account in the system and one of three fixed roles: customer, operator or administrator.
Customer	User who registers vehicles, books and pays for services, and rates the service received.
Operator	Car wash employee who performs the assigned services and records their start and finish.
Administrator	Person in charge of the car wash operation; configures the system and reviews payments.
Booking	Confirmed time slot to wash one of the customer's vehicles on a date, at a time and in a bay. It contains one or more service lines.
Service line	Each service included in a booking, with the price in force when the booking was made.
Bay	Physical wash station at the site. Two active bookings cannot use the same bay at overlapping times.
Time slot	Start interval offered by the system when booking; 30 minutes by default.
Opening hours and exception	Weekly opening hours of the site and dates with a special opening or closing (for example, public holidays).
Operator shift	Weekly schedule in which the operator can receive assignments.
Absence	Period in which an operator cannot receive assignments.
Operator profile	The operator's specialty and skills, each skill with its level (certified, expert or advanced).
Inspection report	Review of the vehicle in five phases (reception, exterior, interior, engine and delivery) with findings and photos, shared with the customer through a link. It applies to services of 180 minutes or more.
Finding	Note in the inspection report about an area of the vehicle, with its severity (informational, minor or major) and up to 5 photos.
Business data	Legal, tax and contact information of the car wash that the administrator keeps in the settings.
Service execution	Record of the work done by an operator on a service line, with its status and times.
Payment method	Cash, Nequi, Daviplata or bank transfer.
Payment account	Car wash details (holder, number, QR code and instructions) used to receive payments through a digital method.
Receipt	JPG or PNG image that the customer uploads as proof of a digital payment.
Reference code	Transaction number shown on the receipt. It cannot be repeated across different bookings.
Rating	Whole score from 1 to 5, with an optional comment, that the customer gives to a completed booking.
Promotion	Coupon with a code, validity period, discount percentage and, optionally, a minimum number of points to unlock it.
Loyalty points	Points the customer earns for each paid service; they unlock promotions and are not deducted when a promotion is redeemed.
Points ledger	Record of point movements; it is the only valid source of the customer's balance.
Notification	Message to a user about a relevant event, with a read or unread status.

Table 3 *Acronyms and Abbreviations*

Acronym	Meaning
ADR	Architecture Decision Record: record of an architecture decision with its context and consequences.
ADSO	Software Analysis and Development Technologist program (Tecnólogo en Análisis y Desarrollo de Software).
AMQP	Advanced Message Queuing Protocol, the messaging protocol used by RabbitMQ.
API / REST API	Application programming interface; the system's API exposes resources over HTTP with JSON bodies.
API Gateway	Single entry point that receives the requests from the web and the app and routes them to the right service.
BCrypt	Adaptive password-derivation function used to store passwords (Provos & Mazières, 1999).
COP	Colombian peso.
CORS	Cross-Origin Resource Sharing, the browser mechanism that authorizes requests between origins.
CU	Use case (*caso de uso*); the identifiers are kept from the Spanish version.
HU	User story (*historia de usuario*).
JWT	JSON Web Token: signed token that identifies the user and their role in each request (Jones et al., 2015).
Liquibase	Tool that versions and applies database schema changes.
Microservice	Independently deployable service with a single responsibility that owns its data schemas.
MoSCoW	Prioritization technique: must have, should have, could have and won't have (Wiegers & Beatty, 2013).
QR	Quick response code used to receive digital payments.
RabbitMQ	Message broker through which the services publish domain events.
RF / RNF	Functional requirement / non-functional requirement (*requisito funcional / no funcional*); the identifiers are kept from the Spanish version.
SENA	Servicio Nacional de Aprendizaje, Colombia's National Learning Service.
SMTP	Simple Mail Transfer Protocol, the email sending protocol.
SRS	Software Requirements Specification.
UML	Unified Modeling Language, used in the use case and state diagrams.

1.4 Reference Documents

Besides the bibliographic references listed at the end, this SRS relies on the project's technical documentation (Guilombo Osorio et al., 2026). Table 4 lists the internal documents it is based on.

Table 4 *Internal Reference Documents*

Document	Location in the `vehicle-w-docs` repository	Use in this SRS
Scope and glossary	`01-context/scope.md`, `01-context/glossary.md`	Scope, exclusions, assumptions and constraints

Document	Location in the `vehicle-w-docs` repository	Use in this SRS
Domain model	`02-domain/entities-and-rules.md`, `02-domain/domain-events.md`	Business rules, life cycles and events
Requirements and user stories	`04-requirements/`	Functional and non-functional requirements, acceptance criteria and traceability
Architecture decisions	`05-architecture/decisions/records/` (ADR-001 to ADR-015)	Design decisions and technical constraints
Data model	`06-data/models.md`	Schemas, tables and integrity rules
API contracts	`07-api/contracts/openapi/`	Software and communication interfaces
UML diagrams	`08-uml/diagrams/source/`	Sequence and state diagrams

1.5 Document Overview

Section 2 presents the project context: problem, rationale, objectives and team. Section 3 describes the product, its users, the constraints and the design decisions. Section 4 specifies the external interfaces. Sections 5 and 6 contain the functional and non-functional requirements, each module with screenshots of the running system. Section 7 develops the use case model (with one diagram per actor) and section 8 the business flows and life cycles. Section 9 defines the data requirements, section 10 the traceability and section 11 the progress status and open decisions.

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2. Project Context

2.1 Problem Statement

How can a digital system organize bookings, operator assignment, payment records and service tracking in a car wash, in order to reduce operational errors and improve the customer experience?

Today, customers only find out whether a time slot is free once they get to the site, bookings are written down by hand and easily overlap, and the operator depends on verbal instructions to know which service is theirs. In addition, a cash payment is not linked to the booking, and there is no history to review the service provided or the customer's rating.

2.2 Rationale

A car wash with digital bookings reduces walk-ins without an appointment, organizes the use of the bays and the operators' workload, and leaves a verifiable record of every payment. For

the customer, knowing the actual availability avoids unnecessary trips; for the administrator, receipt review and operational reports replace manual control. For the team, the project applies what was learned in the program: microservice architecture, domain-driven design (Evans, 2003), hexagonal architecture (Cockburn, 2005), concurrency control, JWT authentication and automated testing.

2.3 Objectives

2.3.1 General Objective

To develop a web and mobile system that organizes the booking, operator assignment, payment and tracking of washes in a car wash, ensuring that each bay and each operator is assigned without schedule overlaps and that each payment is linked to its booking.

2.3.2 Specific Objectives

- Allow the customer to register their vehicles and book a service at an available time from the web or the Android app.
- Validate the site's opening hours and bay availability before confirming a booking, and offer alternative times when the slot is taken.
- Assign operators without schedule overlaps, blocking anyone who is off shift, absent or already has a booking at that time.
- Record payments by QR code, bank transfer or cash, with receipt review by the administrator.
- Build an administration module to manage the catalog, prices, opening hours, bays, operators, payments, promotions and reports.
- Allow the customer to rate each completed service and earn points that unlock promotions.

2.4 People Involved

Table 5 *Project Team*

Name	Role	Responsibility	Contact
Brayan Estiven Patiño Cabrera	Project lead	Coordinates development, manages scope and schedule, and makes sure the requirements are met.	bspc1507@gmail.com
Merari Urbano Quintero	Backend lead	Develops the business logic of the services and exposes it through the REST API.	merariurbano0322@gmail.com
Juan Camilo Guilombo Osorio	Frontend lead	Designs and develops the web and mobile interfaces, and keeps the visual style consistent across platforms.	cguilombo587@gmail.com

Note. All three members are apprentices of SENA's Software Analysis and Development Technologist program, group 3145556.

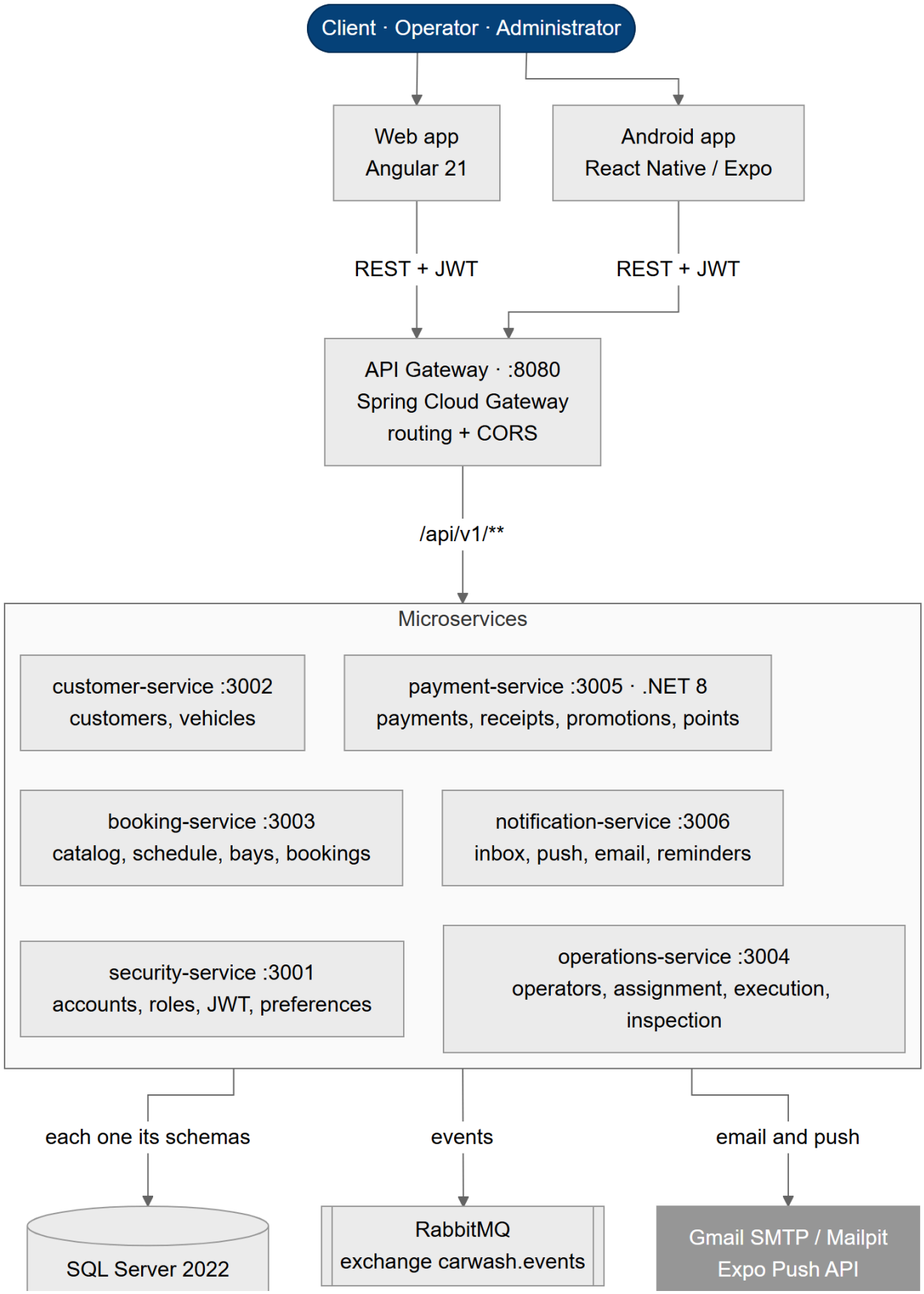
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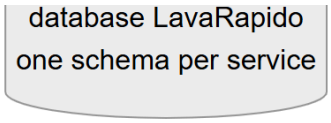
3. Overall Description

3.1 Product Perspective

LavaRápido is a new product that replaces the manual coordination of a car wash. It does not integrate with a payment gateway: digital payment happens outside the platform, by QR code or bank transfer, and the system receives the receipt, puts it under review and records the administrator's decision (ADR-008). Figure 1 shows the components that were built and how they actually communicate.

Figure 1 *LavaRápido Component Architecture (C4, Container Level)*


Client · Operator · Administrator
Web app
Angular 21
Android app
React Native / Expo
REST + JWT
REST + JWT
API Gateway · :8080
Spring Cloud Gateway
routing + CORS
/api/v1/**
Microservices
customer-service :3002
customers, vehicles
payment-service :3005 · .NET 8
payments, receipts, promotions, points
booking-service :3003
catalog, schedule, bays, bookings
notification-service :3006
inbox, push, email, reminders
security-service :3001
accounts, roles, JWT, preferences
operations-service :3004
operators, assignment, execution, inspection
each one its schemas
events
email and push
SQL Server 2022
RabbitMQ
exchange carwash.events
Gmail SMTP / Mailpit
Expo Push API



Note. All services share one SQL Server instance, but each one owns its schemas and there are no foreign keys between services (ADR-003). Table 55 links each event to its effect and section 3.3 of the *Software Design Documentation* details the REST calls between services. Diagram by the authors, based on `lavarapido-infra/docker-compose.yml`.

Table 6 *System Components and Their Responsibility*

Component	Port	Technology	Responsibility
API Gateway	8080	Spring Cloud Gateway	Single entry point for the web and the app; routes by path and answers CORS. It does not expose internal routes.
security-service	3001	Java 21, Spring Boot 4	Accounts, roles and permissions, JWT authentication, lockout after failed attempts, password recovery, profile address and preferences.
customer-service	3002	Java 21, Spring Boot 4	Customer profile, vehicles and vehicle types.
booking-service	3003	Java 21, Spring Boot 4	Catalog and prices, opening hours, bays, availability, bookings, day summary and business data.
operations-service	3004	Java 21, Spring Boot 4	Operators, profile, shifts, absences, automatic and manual assignment, execution, ratings, metrics and inspection report.
payment-service	3005	.NET 8	Payment accounts, payments, receipts, history with filters, refunds, promotions and points.
notification-service	3006	Java 21, Spring Boot 4	Notification inbox, push, emails, reminders and administrator alerts, in each account's language.
Web application	—	Angular 21	Interface for the three roles in the browser.
Mobile application	—	React Native 0.86, Expo SDK 57	Interface for the three roles on Android.
SQL Server	1433	SQL Server 2022	Database: 9 schemas and 57 tables, versioned with Liquibase.
RabbitMQ	5672	RabbitMQ 4	Domain event bus (exchange `carwash.events`).

3.2 Product Functions

The functions are grouped into nine modules, which organize the functional requirements of section 5 (see Table 7).

Table 7 *Functional Modules of the System*

Module	Main functions	Requirements
Authentication and accounts	Registration, sign-in, password recovery and change, lockout, own account, users, roles and permissions	RF-001 to RF-004b, RF-027 to RF-029
Vehicles	Registration and management of the customer's vehicles	RF-030
Catalog, opening hours and bays	Services and prices by vehicle type, opening hours with exceptions, bays and business data	RF-021, RF-031, RF-032, RF-037
Bookings	Request, availability, rescheduling, cancellation and administrative management	RF-005 to RF-007, RF-033
Operations	Assignment, assigned services, execution, rating, availability, operator profile and inspection report	RF-008 to RF-012b, RF-022, RF-036, RF-038
Payments	Payment accounts, methods, receipts, review, history, refunds	RF-013 to RF-016, RF-034, RF-035
Dashboard and notifications	Dashboard per role, notifications, theme and language	RF-017 to RF-020
Reports	Services, revenue, ratings and performance	RF-023
Promotions and points	Promotions, coupon redemption and loyalty points	RF-024 to RF-026

3.3 User Characteristics

The system has three fixed roles and one non-human actor (see Table 8). No role requires technical training; knowing how to use a browser or a mobile app is enough.

Table 8 *System Actors*

Actor	Description	Main activities
Customer	Person who brings their vehicle to the car wash. Uses the Android app or the web.	Registers vehicles, books (and may choose the operator), cancels, pays, applies promotions, checks their history, rates the service and receives the inspection report.
Operator	Employee who washes the vehicles. Uses the app or the web at the site.	Checks their assigned services, schedule and ratings, and indicates when each service starts and ends.
Administrator	Person in charge of the car wash operation.	Sets up the catalog, prices, opening hours, bays, payment accounts and business data; manages users, operators and bookings; assigns or reassigns services; records the inspection report; reviews payments; manages promotions; checks reports and metrics.
System	Automatic processes (timers and event consumers).	Validates availability, assigns operators when the booking is confirmed, sends reminders, alerts and notifications, and credits points.

3.4 Constraints

Table 9 *System Constraints*

ID	Type	Constraint
RT-01	Technical	The backend uses Java 21 with Spring Boot 4 and Spring Cloud Gateway; payment-service uses .NET 8 (ADR-013). The web uses Angular 21 and the mobile app uses React Native with Expo.
RT-02	Technical	Every external request enters through the API Gateway (port 8080). The `/internal/**` routes can only be reached between services, with a shared key.
RT-03	Technical	The services use a single SQL Server 2022 instance, with one schema per service and no foreign keys between services (ADR-003).
RT-04	Technical	Business rules are implemented in the domain, with no dependency on frameworks or on the database (hexagonal architecture, ADR-007).
RT-05	Technical	The whole solution runs in Docker containers with Docker Compose, locally and on an Azure virtual machine behind Caddy (HTTPS); no managed cloud services are used.
RD-01	Data	Statuses are stored as English codes and translated in the interface (ADR-010).
RD-02	Data	Records with history (users, vehicles, services, bays) are deactivated, not deleted.
RD-03	Data	The receipt is stored in the database as a data URL and the inspection report photos as binary data, until file storage is available; the server limits the request body to 8 MB.
RD-04	Data	Amounts are expressed in Colombian pesos (COP).
RO-01	Operational	The system runs a single car wash: it does not handle branches, coverage zones or routes.
RO-02	Operational	An internet connection is required to check availability, book and pay.
RO-03	Operational	Payment verification is manual: it depends on the administrator reviewing the receipts.
RO-04	Operational	Assignment depends on the operators' shifts and absences being up to date.
RO-05	Operational	RabbitMQ events require messaging to be enabled (`MESSAGING_ENABLED` or `app.messaging.enabled` flag); it is disabled by default. Automatic assignment also needs the internal key (`INTERNAL_API_KEY`) and can be turned off with `AUTO_ASSIGN_ENABLED=false`.
RR-01	Regulatory	Personal data is handled according to Law 1581 of 2012 (Congreso de la República de Colombia, 2012); the customer accepts the policy when registering, and the server stores the date and time of the acceptance.
RR-02	Regulatory	The system does not capture, transmit or store payment card data.
RP-01	Project	Team of three part-time apprentices; no assigned budget (estimated cost under real conditions: COP 212,925,240, according to the *Technical and Economic Proposal*); delivery on the academic date.

3.5 Assumptions and Dependencies

- Customers accept paying by QR code or bank transfer and uploading the receipt, instead of using an automatic gateway.

- Customers have an Android device or a browser, and an internet connection when booking.
- The site has fixed opening hours and operators with defined shifts; the administrator keeps opening hours, catalog, prices and payment accounts up to date.
- The data volume corresponds to an academic project, not to a large-scale real operation.
- Email depends on an SMTP server (Mailpit in development) and push on the Expo Push service (Expo, n.d.). If they fail, the notification stays in the inbox and the rest of the system keeps working.
- The status of services and ratings is updated when the user opens the screen; there are no real-time updates.

3.6 Design Decisions

During analysis and development, decisions were made that change the original specification. Each one is recorded in an ADR in the documentation repository; Table 10 summarizes those that affect the requirements.

Table 10 *Architecture Decisions That Affect the Requirements*

ADR	Decision	Effect on the requirements
ADR-003	One schema per service in a shared instance, with no foreign keys between services.	References between services are validated through the API (RT-03).
ADR-004	REST for synchronous queries and RabbitMQ for non-blocking notices.	A booking is confirmed even if the notice email fails (RNF-012).
ADR-006	JWT authentication and role-based authorization.	Defines RNF-004 and the redirect by role (RF-002).
ADR-007	Hexagonal architecture in each service.	Business rules are tested without infrastructure (RNF-006).
ADR-008	Manual payment verified with a receipt; no gateway.	Gives rise to RF-013 to RF-016 and RF-034.
ADR-010	Data model aligned with the screens; statuses in English.	Registration does not ask for an address (RF-001); the address is an optional profile field (RF-027) and the business data is stored on the server (RF-037).
ADR-011	Event-driven notifications, with push and email, in each account's language.	Defines the types in RF-018, the alert for bookings without an operator and the inspection report notice.
ADR-013	payment-service in .NET 8.	Technical constraint RT-01.

ADR	Decision	Effect on the requirements
ADR-014	Integration of operations with bookings, and assignment rules.	Defines RF-008, RF-009, RF-010 and RF-012b, with the 15-minute start window.
ADR-015	Points that unlock promotions, a real percentage discount, fixed roles with editable permissions, and a bay per booking.	Redefines RF-024 to RF-026 and gives rise to RF-029.

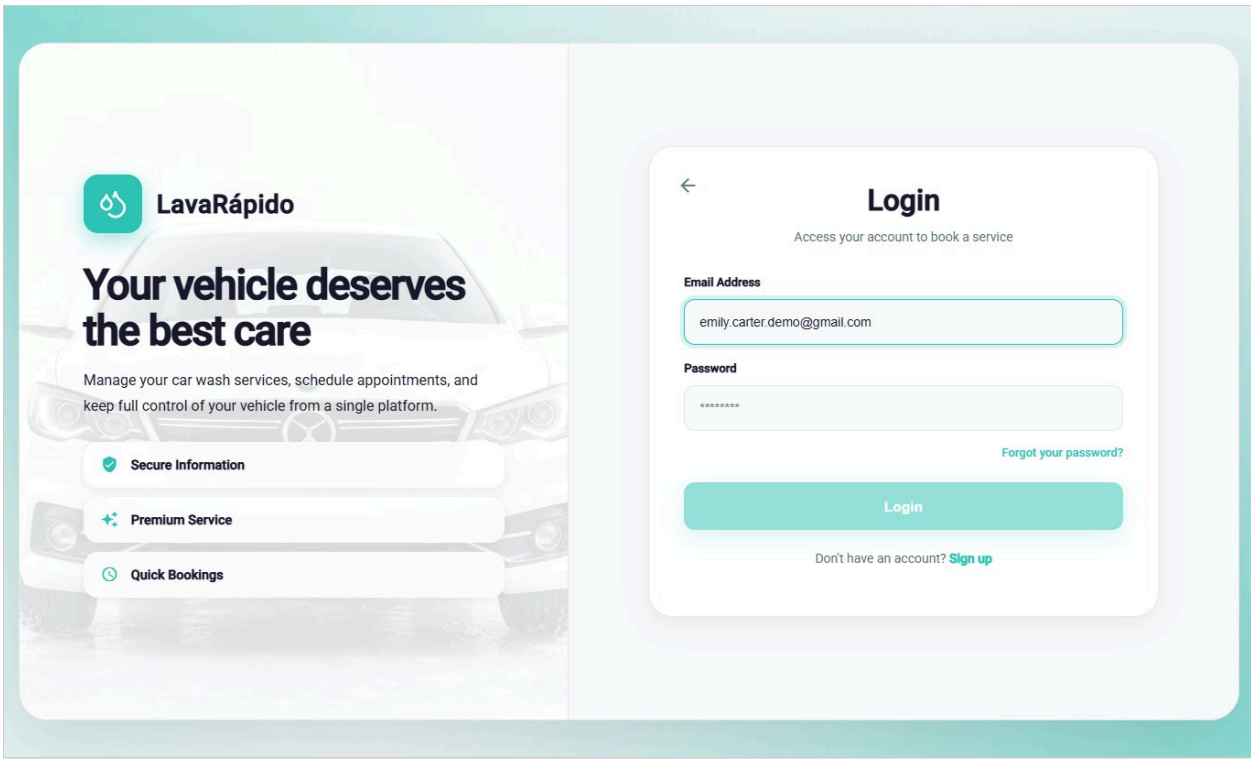
4. External Interface Requirements

4.1 User Interfaces

- The interface is available in Spanish, English, French and Portuguese; amounts are shown in COP and dates according to the selected language.
- Each form points out the field that failed validation, and each action that changes data shows a confirmation or error message.
- Destructive actions (cancelling a booking, deactivating an account, rejecting a payment) ask for confirmation.
- Customer: book (with an optional operator choice), check the history, pay, apply promotions, rate and open the inspection report. Operator: view, start and finish their services. Administrator: management of the catalog, opening hours, bays, business data, users, operators, bookings, inspection reports, payments, promotions and reports.
- There are four color palettes for the user to choose from, saved per account.

Figure 2 shows the actual sign-in screen of the web application, the common entry point for the three roles.

Figure 2 *Sign-In Screen (evidence of RF-002)*



4.2 Hardware Interfaces

The system does not control any hardware devices. Users need a computer with a browser, or an Android phone with a camera or gallery for the receipt. Development requires a computer with Docker Desktop, Java 21, .NET 8 and Node.js.

4.3 Software Interfaces

Table 11 *Software Interfaces*

External system	Version	Purpose	Interface type
Android	8.0 or later	Operating system of the mobile app	Platform
Chrome, Firefox and Edge	Current stable version	Running the web application	Platform
SQL Server	2022	Data persistence for each service	TDS protocol (port 1433)
RabbitMQ	4	Publishing and consuming domain events	AMQP (port 5672)
SMTP server	Mailpit in development	Recovery code, welcome, booking notices and reminders	SMTP with STARTTLS
Expo Push	Expo service	Push notifications to the Android app	HTTPS

4.4 Communication Interfaces

- The web and the app communicate with the API Gateway over HTTPS in every environment reachable from outside the development team's computers; HTTP is only allowed in local development.
- The API uses REST with JSON bodies under the `/api/v1` prefix and receives the token in the `Authorization: Bearer` header.
- Each request carries a correlation identifier (`X-Correlation-Id`) that is written to the logs.
- The services call each other over REST on the `/internal/**` routes, authenticated with the `X-Internal-Key` header, and publish events to the `carwash.events` exchange in RabbitMQ.

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5. Functional Requirements

5.1 Conventions

Each requirement has a unique identifier that is kept across versions. Each requirement is written as a verifiable capability and comes with measurable acceptance criteria (Wiegers & Beatty, 2013).

Priority follows the MoSCoW technique: Essential (must have), Desirable (should have) and Optional (could have). The status of each requirement was verified against the code and the automated tests on October 20, 2026: Implemented (built and tested), Partial (partly built; what is missing is stated) and Pending (not built).

5.2 Authentication and Accounts

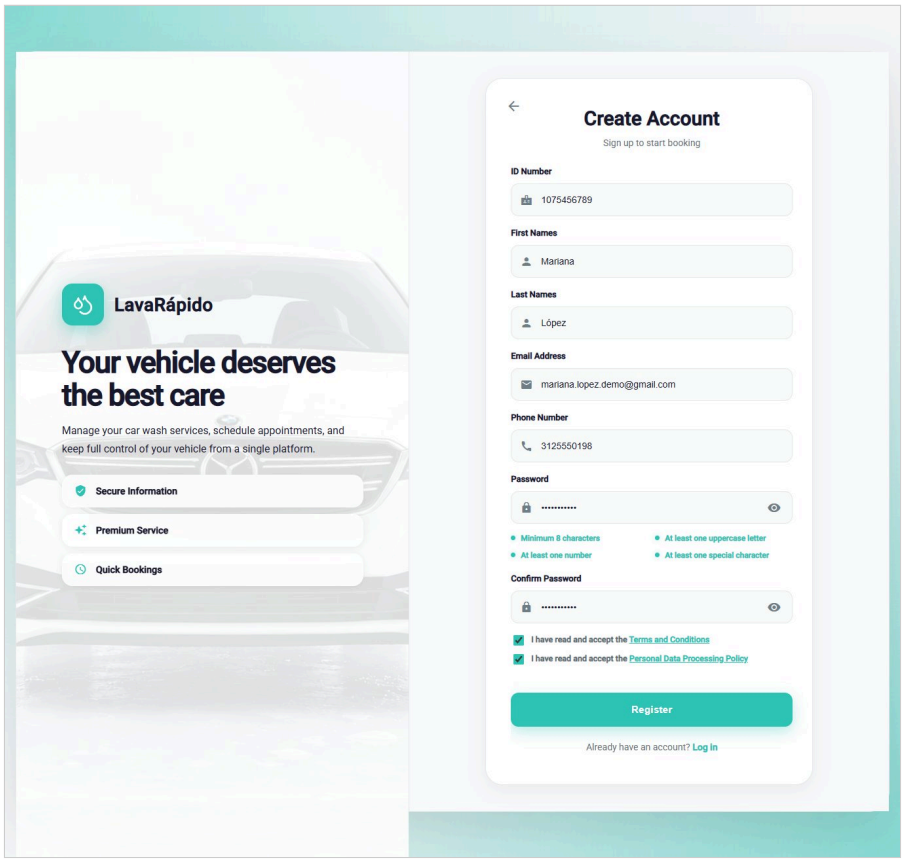
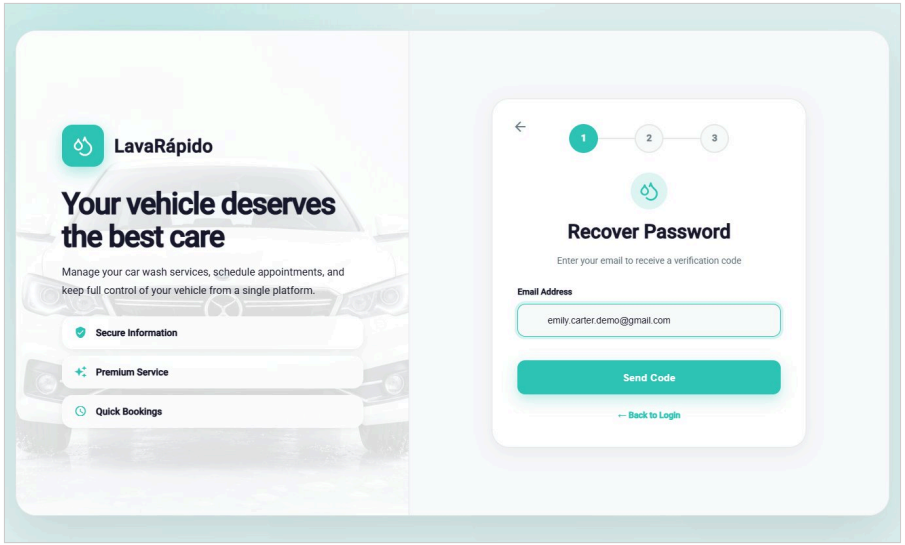
Table 12 *Authentication and Account Requirements (security-service)*

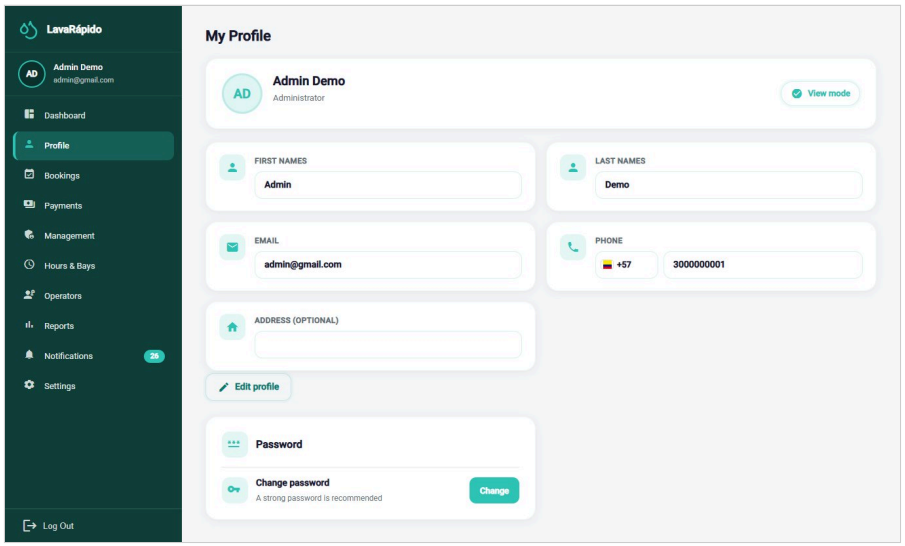
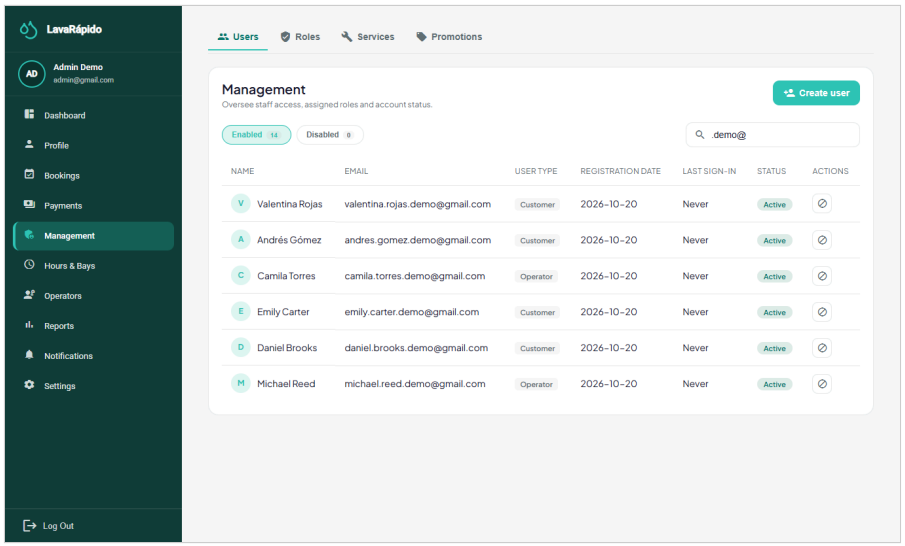
ID	Requirement	Acceptance criteria	Priority	Status
RF-001	**Customer registration.** The system must allow a person to create a customer account with ID number, first names, last names, phone, email and password, after accepting the personal data processing policy.	• If the email is already registered, the system rejects the request with a message saying so and does not create the account. • The password meets the policy in RNF-004. • When the account is created, the customer profile is created in customer-service and a welcome email is sent. • No address is requested, because the service is provided at the site.	Essential	Implemented

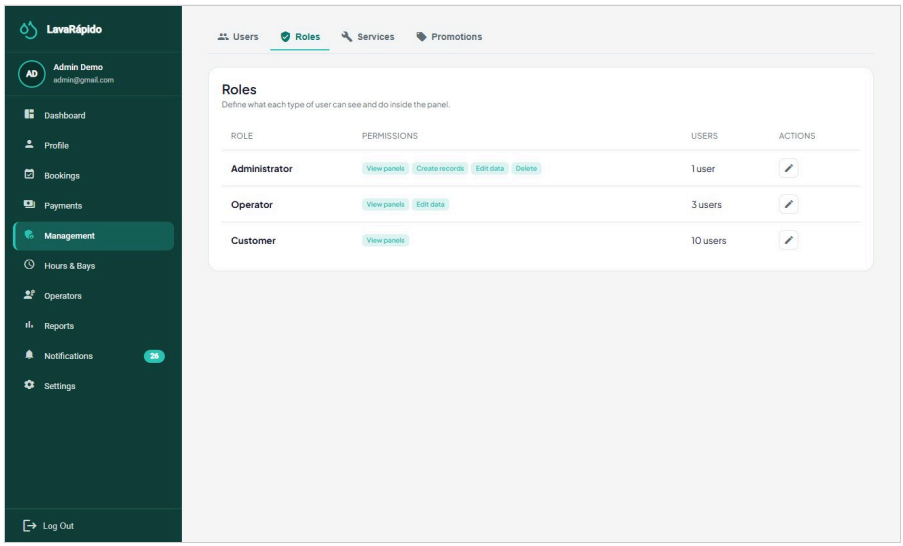
ID	Requirement	Acceptance criteria	Priority	Status
RF-002	**Sign-in.** The system must authenticate the user with email and password, issue a JWT token and redirect them to their role's dashboard.	• With valid credentials, the user reaches their role's dashboard (customer, operator or administrator). • With invalid credentials, the message is generic and does not reveal whether the email exists. • A deactivated account cannot sign in.	Essential	Implemented
RF-003	**Password recovery.** The system must allow resetting the password with a 6-digit code sent to the registered email.	• The code expires after 15 minutes and is invalidated after 5 failed attempts. • For an unregistered email, the response is identical and no email is sent. • The code is stored only as a hash, is single-use, and requesting a new one invalidates the previous one. • The new password cannot be the same as the current one.	Desirable	Implemented
RF-004	**Password change.** The system must allow the authenticated user to change their password after validating the current one.	• If the current password does not match, the change is rejected and the password stays the same. • The new password meets the policy in RNF-004.	Desirable	Implemented
RF-004b	**Lockout after failed attempts.** The system must lock the account for 15 minutes after 5 failed sign-in attempts within 15 minutes.	• During the lockout, sign-in is rejected with a 423 response (`ACCOUNT_LOCKED`) and a message explaining the lockout, even if the password is correct. • A correct sign-in before the fifth failure clears the count. • A successful password reset (RF-003) lifts the lockout.	Essential	Implemented
RF-027	**Own account management.** The system must allow the user to view and update their profile data and contact address, change their email, sign out and deactivate their account.	• Signing out revokes the session on the server. • Deactivating the account asks for the current password and closes every session; an administrator cannot close their own account. • The address is optional, has up to 200 characters and is stored on the server, so it is the same on the web and in the app; saving it empty deletes it. • Accounts are deactivated, not deleted, to keep bookings, payments and ratings.	Desirable	Implemented

ID	Requirement	Acceptance criteria	Priority	Status
RF-028	**User administration.** The system must allow the administrator to list users with their registration date, create operator and administrator accounts, and activate or deactivate accounts.	• A deactivated account cannot sign in, all its sessions are closed and it keeps its history. • The role is set when the account is created (customer through public registration; operator or administrator when an administrator creates it) and the system does not offer to change it afterwards. • An administrator account cannot be deactivated (`ADMIN_CANNOT_BE_DISABLED`) and the administrator cannot deactivate their own account. • Only the administrator role can use this function (403 response for the other roles).	Desirable	Implemented
RF-029	**Roles and permissions.** The system must handle three fixed roles (administrator, operator and customer) and allow the administrator to view and edit the permission matrix of each role.	• Roles cannot be created or deleted. • In this version the matrix is informational: effective authorization is based on the role in the JWT token (ADR-015).	Optional	Implemented

Evidence in the system (authentication and accounts module)

RF	Screen
RF-001 (registration)	
RF-003 (password recovery)	

RF	Screen
RF-004, RF-027 (profile, password change and own account)	
RF-028 (user administration)	

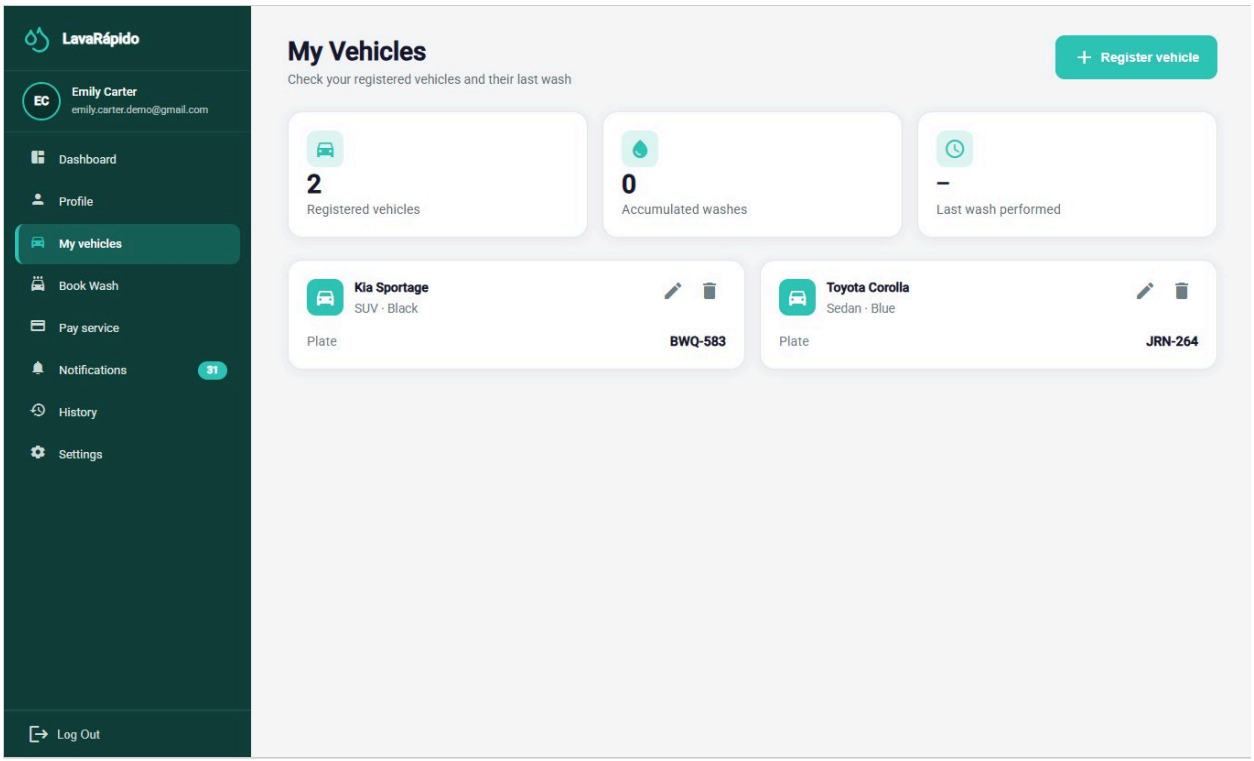
RF	Screen
RF-029 (roles and permissions)	

5.3 Vehicles

Table 13 *Vehicle Requirements (customer-service)*

ID	Requirement	Acceptance criteria	Priority	Status
RF-030	**Customer vehicle management.** The system must allow the customer to register, view, edit and remove their vehicles, indicating the license plate and vehicle type.	• Only the owner manages their vehicles; the administrator can view them. • Removing a vehicle does not delete the past bookings in which it was used. • The vehicle type determines the price of each service (RF-031).	Essential	Implemented

Evidence:



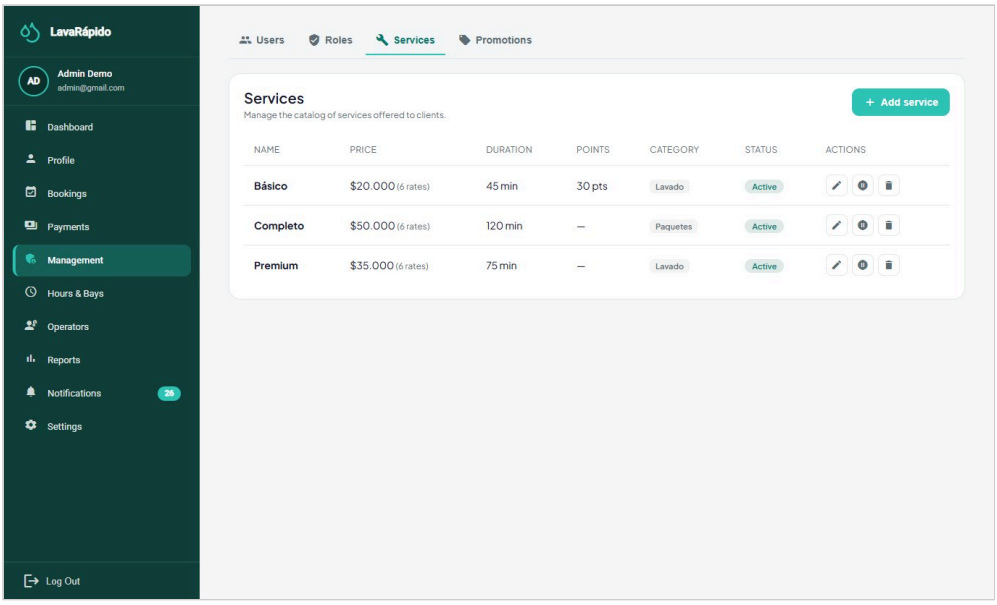
5.4 Catalog, Opening Hours and Bays

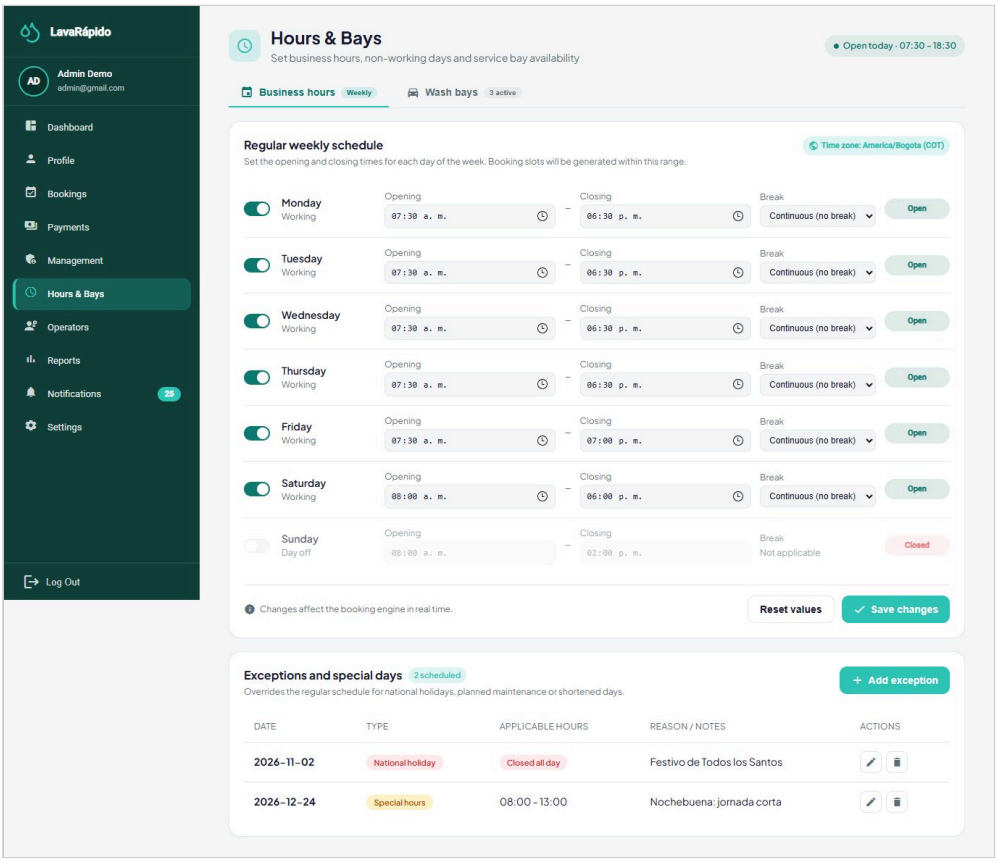
Table 14 *Catalog, Opening Hours and Bay Requirements (booking-service)*

ID	Requirement	Acceptance criteria	Priority	Status
RF-031	**Service catalog and prices.** The system must allow the administrator to create, edit, activate, deactivate and delete wash services, with category, description, price and estimated duration by vehicle type, and loyalty points.	• A price change does not modify existing bookings: each line keeps the price in force when the booking was made. • Two overlapping active rates are not allowed for the same service and vehicle type. • Only active services are offered when booking.	Essential	Implemented
RF-021	**Opening hours and exceptions.** The system must allow the administrator to set the site's weekly opening hours, the daily break and the exceptions (public holidays and special hours).	• If the closing time is earlier than or equal to the opening time, the change is rejected. • The new opening hours apply to later availability checks. • Each change is recorded in the opening-hours history.	Desirable	Implemented

ID	Requirement	Acceptance criteria	Priority	Status
RF-032	**Bay management.** The system must allow the administrator to create, edit, deactivate and delete the site's wash bays.	• Only active bays are used to confirm new bookings.	Essential	Implemented
RF-037	**Business data.** The system must allow the administrator to view and update the car wash data: legal information (legal name, tax ID, business type, foundation date, legal representative), tax information (tax regime, CIIU activity, DIAN resolution, invoice prefix and range) and service channels (website, social networks, support line and service hours).	• The data is stored in booking-service and is the same on the web and in the app. • The legal name, tax ID and address are required; the phone has at least 7 digits, the email must be valid and the foundation date cannot be in the future. • The public page and the help center show the saved email, phone, support line and service hours.	Desirable	Implemented

Evidence:

RF	Screen
RF-031 (catalog and prices)	

RF	Screen
RF-021, RF-032 (opening hours and bays)	

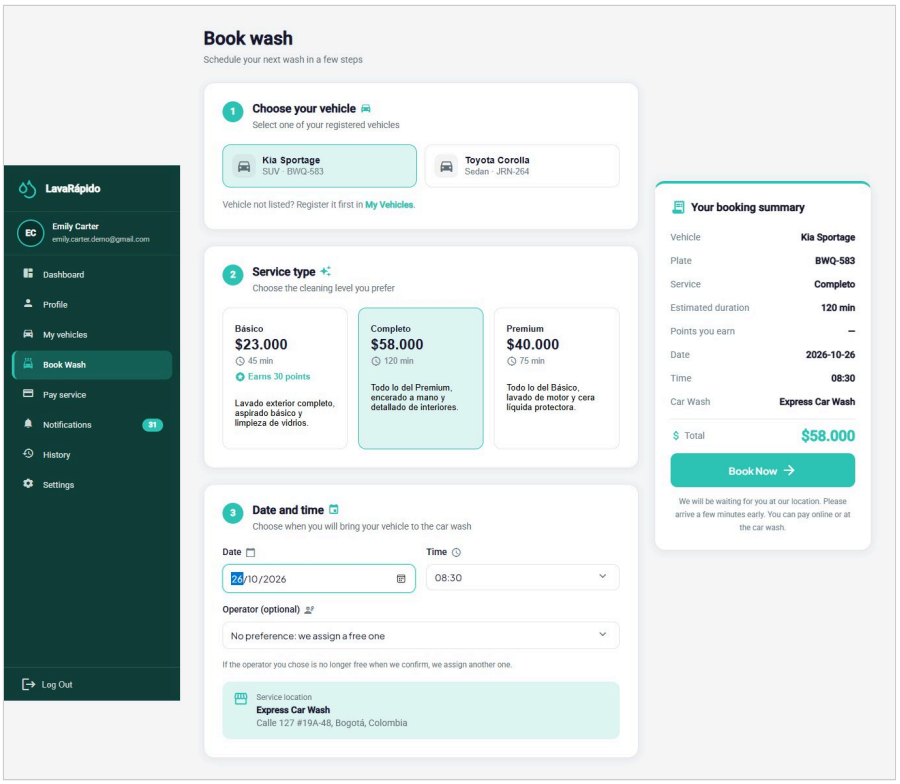
5.5 Bookings

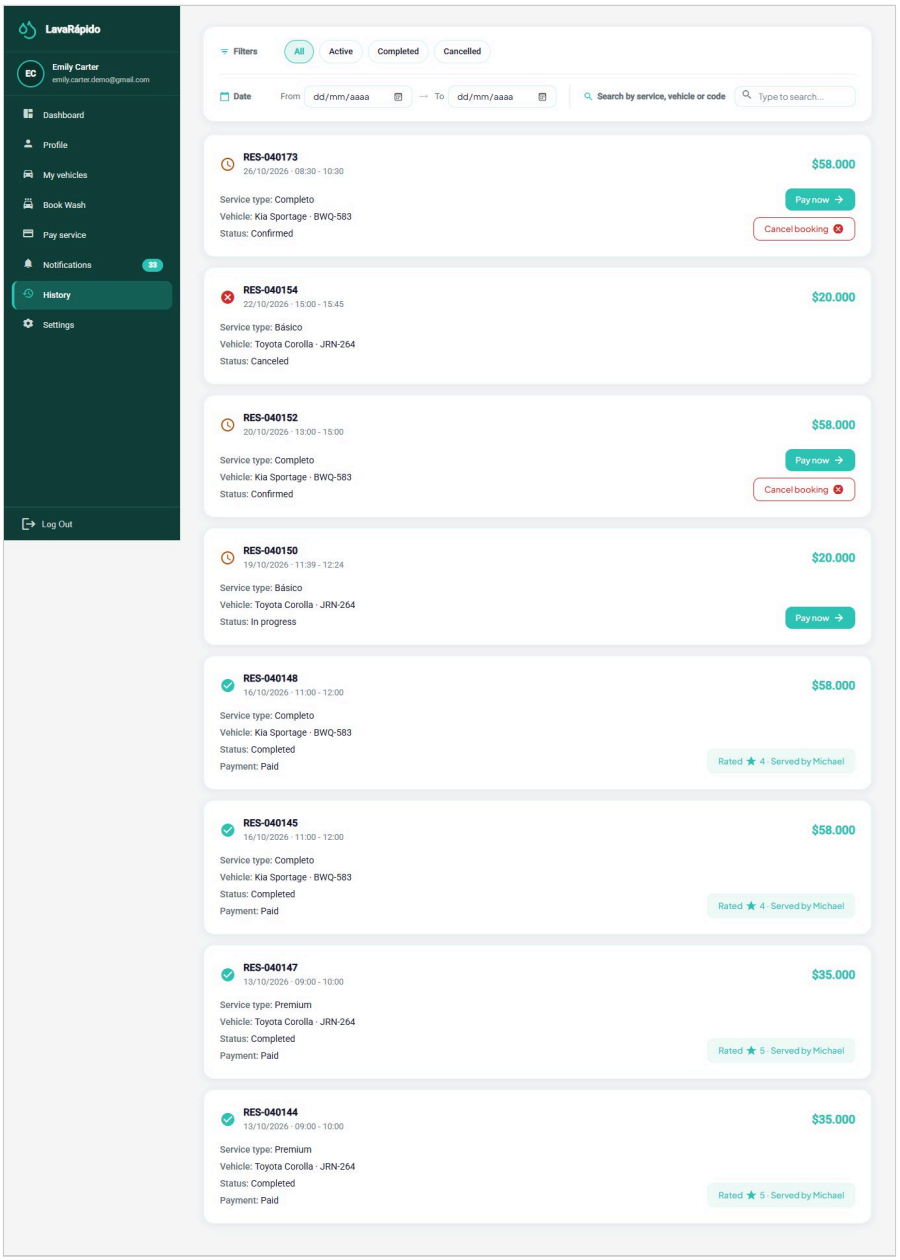
Table 15 *Booking Requirements (booking-service)*

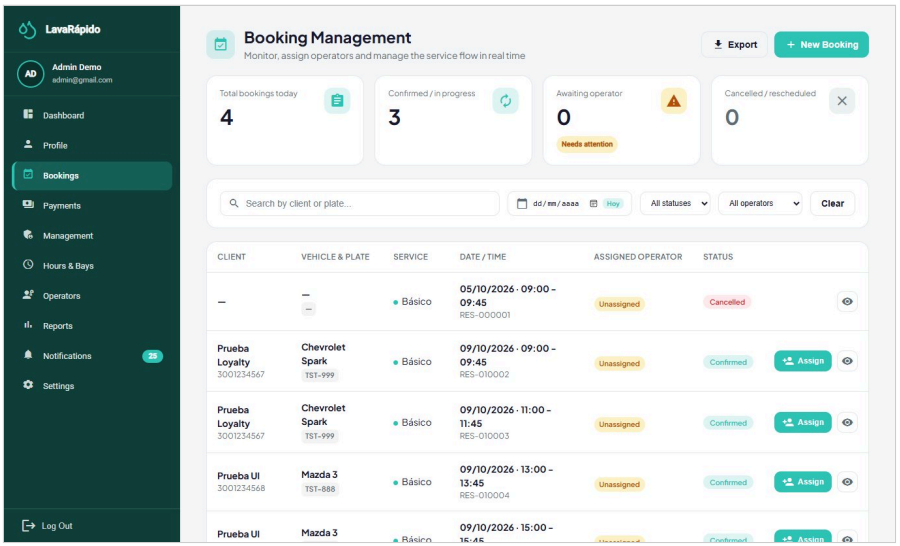
ID	Requirement	Acceptance criteria	Priority	Status
RF-005	**Booking request.** The system must allow the customer to book one or more services for one of their vehicles, choosing the date and start time.	• The price of each service matches the selected vehicle type. • Times are offered in 30-minute slots and up to 60 days ahead (configurable values). • Only the customer's own vehicles are accepted.	Essential	Implemented

ID	Requirement	Acceptance criteria	Priority	Status
RF-006	**Availability check.** Before confirming, the system must validate the site's opening hours, their exceptions and the existence of an active bay without overlapping bookings.	• With a free bay, the booking is confirmed immediately with the bay assigned. • With no free bay, nothing is saved and up to 5 alternative times on the same day are offered. • When two requests arrive at the same time for the same bay and time, only one is confirmed (RNF-003).	Essential	Implemented
RF-007	**Rescheduling and cancellation.** The system must allow the customer to cancel their booking while it is in the Scheduled or Confirmed status and before its start time, and the administrator to reschedule it (RF-033).	• The customer cancels from their history by confirming a dialog, without writing a reason: the system stores the reason `CUSTOMER_REQUEST` and frees the bay. • Rescheduling, done by the administrator, assigns a new time and bay, and keeps the prices. • Once the service is in progress or finished, or, for the customer, once the start time has passed, the action is blocked (`BOOKING_NOT_CHANGEABLE`). • The API also lets the customer reschedule (`PUT /bookings/{id}`), but the web and the app do not offer that screen (improvement action AM-03).	Essential	Implemented
RF-033	**Administrative booking management.** The system must allow the administrator to view all bookings within a date range and change their status (confirm, cancel with a reason, mark as "no-show") or reschedule them, following the life cycle in Figure 4.	• A transition that the life cycle does not allow is rejected.	Desirable	Implemented

Evidence:

RF	Screen
RF-005, RF-006 (booking request)	

RF	Screen
RF-007 (history and cancellation)	

RF	Screen
RF-033, RF-009 (administrative management and assignment)	

5.6 Operations

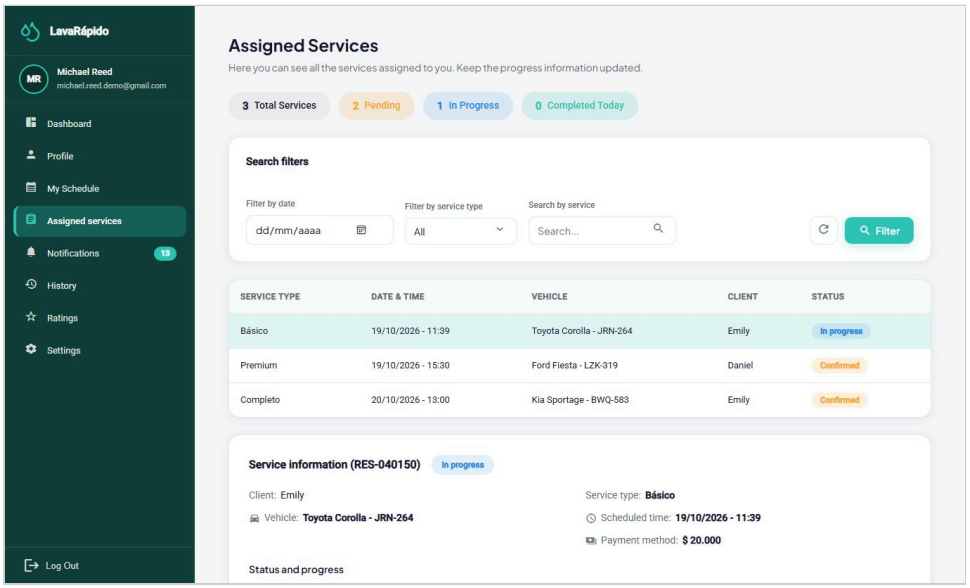
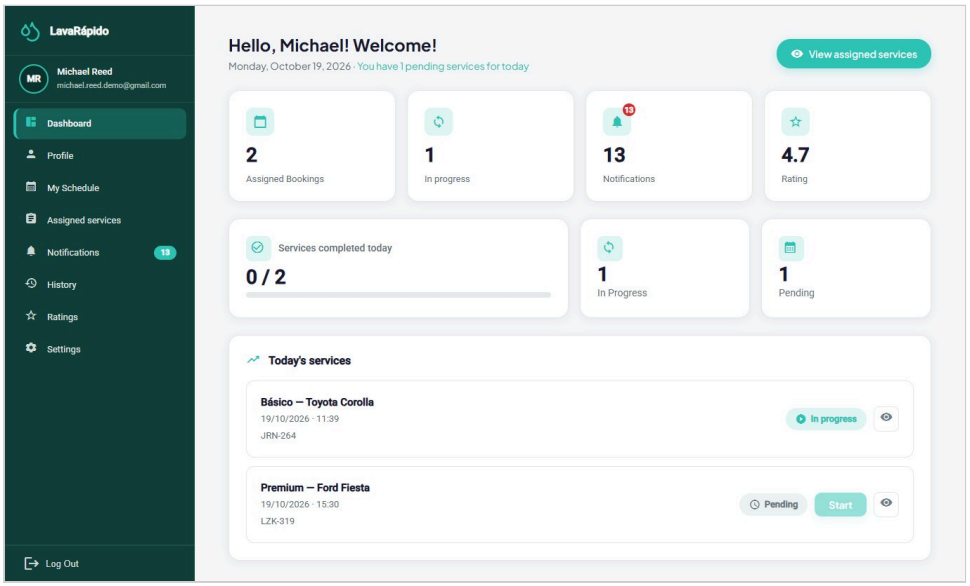
Table 16 *Operations Requirements (operations-service)*

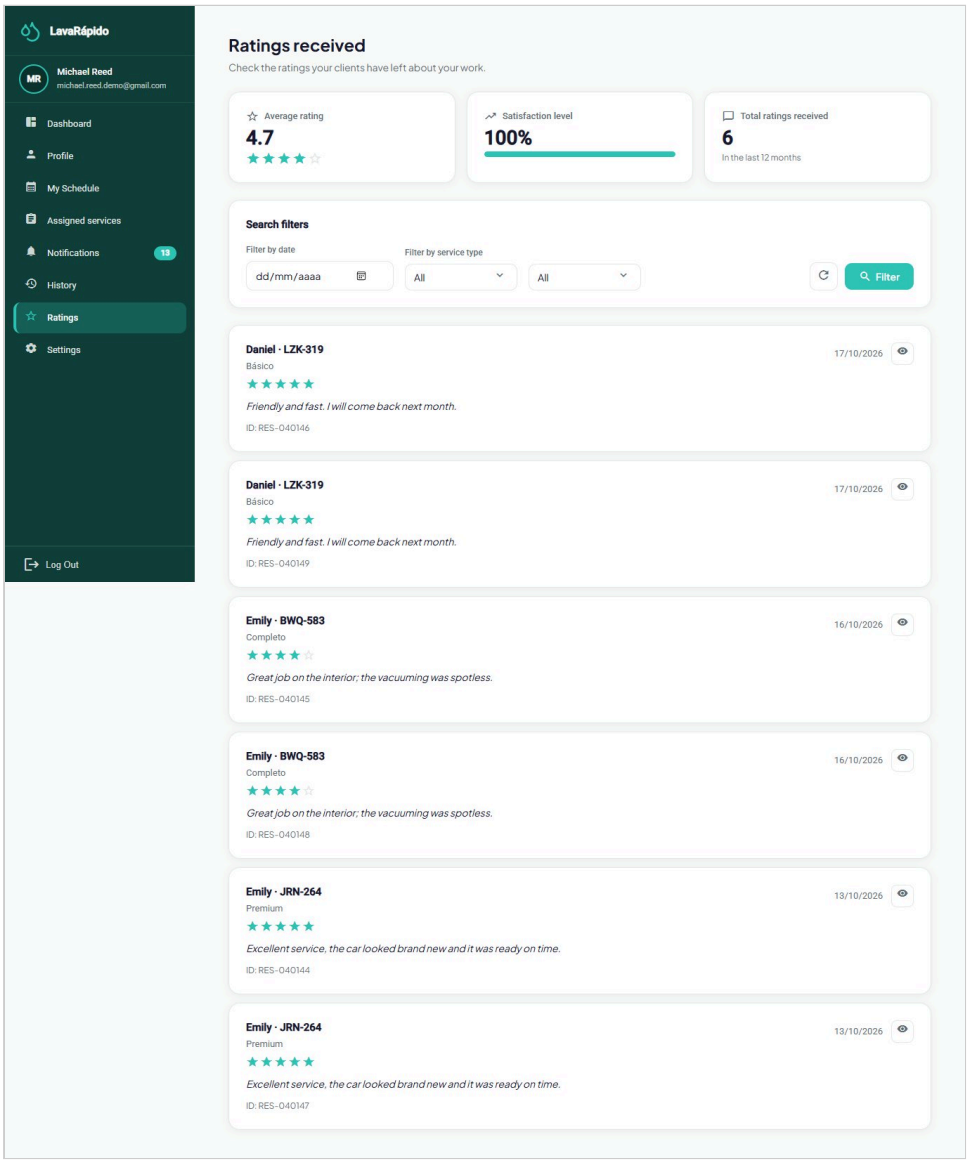
ID	Requirement	Acceptance criteria	Priority	Status
RF-008	**Automatic operator assignment.** The system must automatically assign an available operator to each confirmed booking.	• An operator is eligible if their account is active, they have a shift that covers the slot, and they have no absence and no other overlapping booking. • If the customer chose an operator (RF-010) who is still free, that operator is assigned; otherwise the one with the fewest bookings that day is chosen, and ties are broken by the lowest identifier. • Rescheduling keeps the operator the booking already had. • If no operator is available, the booking stays unassigned and the administrators get an alert when less than 2 hours remain before the start, so they can assign one manually (RF-009).	Essential	Implemented

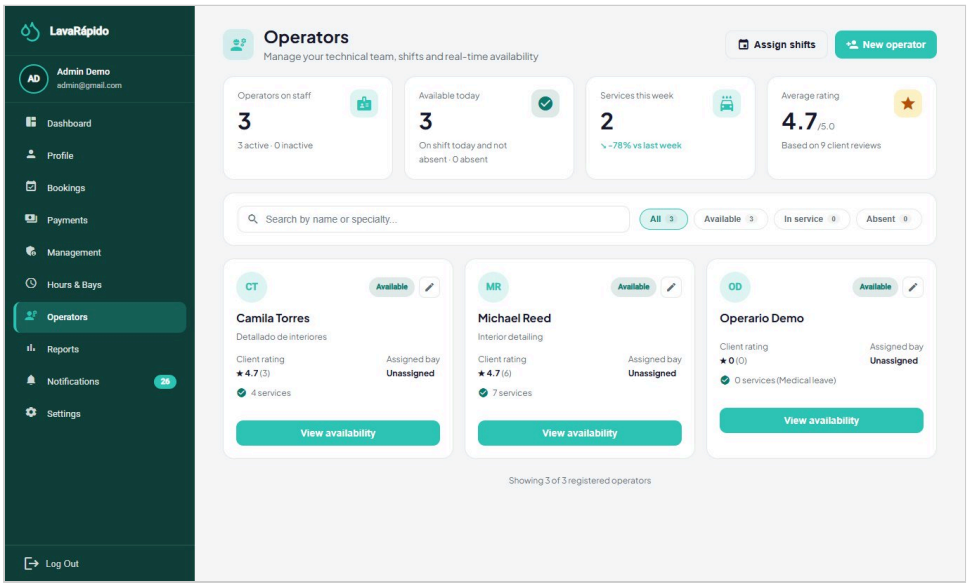
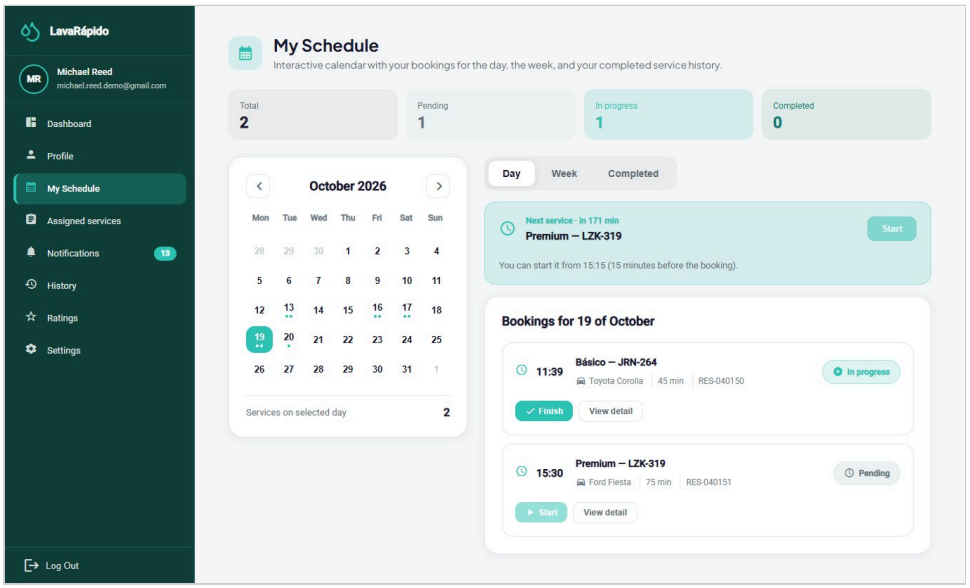
ID	Requirement	Acceptance criteria	Priority	Status
RF-009	**Manual assignment and reassignment.** The system must allow the administrator to assign or reassign an operator to a booking from a list of candidates.	• The list shows available operators first, sorted by name, with their average rating, and the reason why the rest are unavailable. • Assigning an operator who is inactive, off shift, absent or has another booking at that time is blocked. • Reassignment is only possible while the execution is pending; the assigned operator receives a notification.	Desirable	Implemented
RF-010	**Operator choice by the customer.** The system must allow the customer to optionally choose an operator from the list of available ones when booking.	• After picking the time, the customer sees the operators who are free at that time, with their first name and average rating; no contact data is shown. • If they choose no one, the car wash assigns a free operator (RF-008). • If the chosen operator is no longer free when the booking is confirmed, another free operator is assigned and the customer sees on their home screen the name of the person who will serve them.	Desirable	Implemented
RF-011	**Assigned services.** The system must show the operator their assigned bookings with customer, vehicle, time, services and bay.	• The operator only sees the bookings assigned to them.	Essential	Implemented
RF-012b	**Service start and finish.** The system must allow the operator to mark the service assigned to them as started and as finished.	• It can only be started from 15 minutes before the booking time; before that, the button is disabled and the server answers `SERVICE_TOO_EARLY`. • The booking moves to In progress when started and to Completed when finished; starting it cancels the pending reminders. • The action is blocked if the booking is assigned to another operator.	Essential	Implemented
RF-012	**Service rating.** The system must allow the customer to rate one of their completed bookings, only once, with a whole score from 1 to 5 and an optional comment of up to 500 characters.	• A booking that is not completed cannot be rated, and a booking cannot be rated twice. • The comment requires a score. • The rating applies to all lines of the booking and feeds the operator's average. • The customer sees the operator's first name and the operator sees the customer's first name.	Desirable	Implemented

ID	Requirement	Acceptance criteria	Priority	Status
RF-022	**Operator availability.** The system must record each operator's weekly shift and absences, and allow activating or deactivating them.	• The administrator records and adjusts shifts and absences; the operator views their schedule. • The recorded availability feeds RF-008, RF-009 and RF-010. • The administrator sees the team summary: active operators, operators available and absent today, services this week against the previous one, and average rating.	Desirable	Implemented
RF-036	**Vehicle inspection report.** The system must allow the administrator to record an inspection report for bookings of 180 minutes or more and share it with the customer.	• The report has five phases, in the order in which the vehicle is walked through (reception, exterior, interior, engine and delivery), each one Pending, In progress or Done; recording a finding in a pending phase moves it to In progress. • Each finding states the area (up to 120 characters), a note (up to 500), the severity (informational, minor or major) and up to 5 photos. • It can only be created for Confirmed, In progress or Completed bookings, and only edited while the booking is Confirmed or In progress. • The link only shows the report after it is published, and only a published report can be shared; when it is shared, the customer gets a notification with the link. • The public link uses a random code, not the booking number, and shows a masked license plate. • The report is managed from the web and from the mobile app; on the phone the photos are taken with the camera or picked from the gallery and are shrunk before uploading.	Desirable	Implemented
RF-038	**Operator profile.** The system must allow the administrator to record the operator's specialty and skills, each with its level (certified, expert or advanced).	• The specialty has up to 80 characters. • An operator has at most 10 skills, each with a unique name of up to 60 characters. • Saving replaces the whole list of skills.	Optional	Implemented

Evidence:

RF	Screen
RF-011 (services assigned to the operator)	
RF-012b (start and finish, operator dashboard)	

RF	Screen
RF-012 (service ratings)	

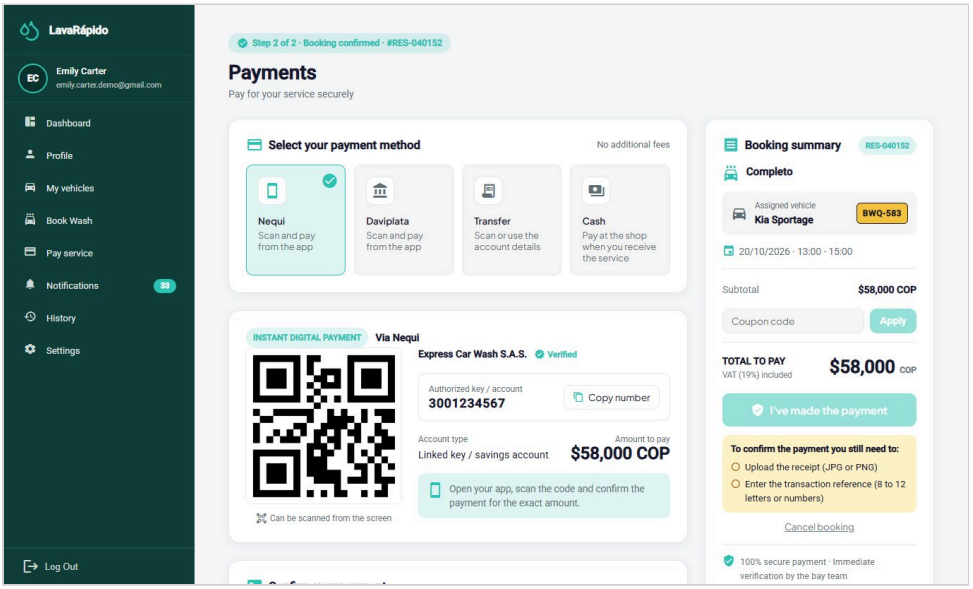
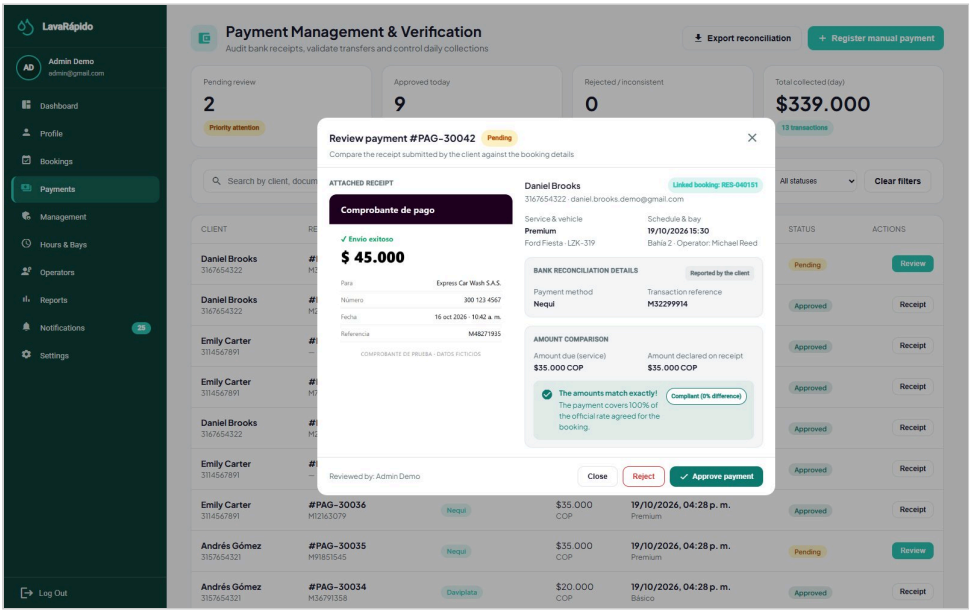
RF	Screen
RF-022 (operator availability, admin view)	
RF-022 (the operator's own view)	

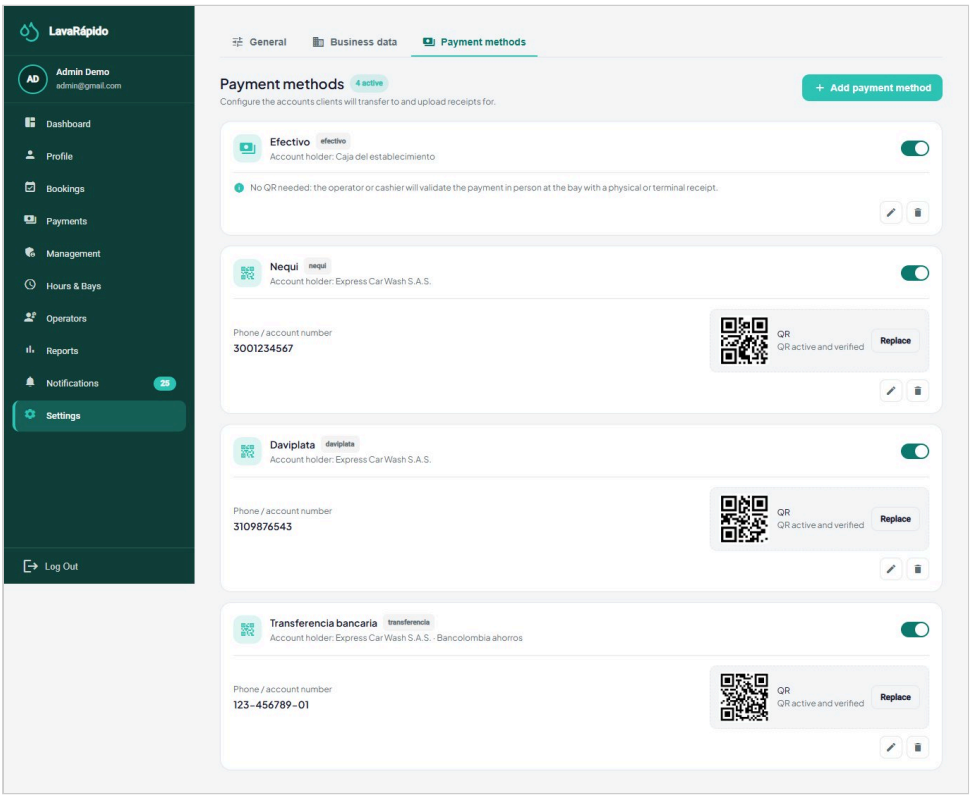
5.7 Payments

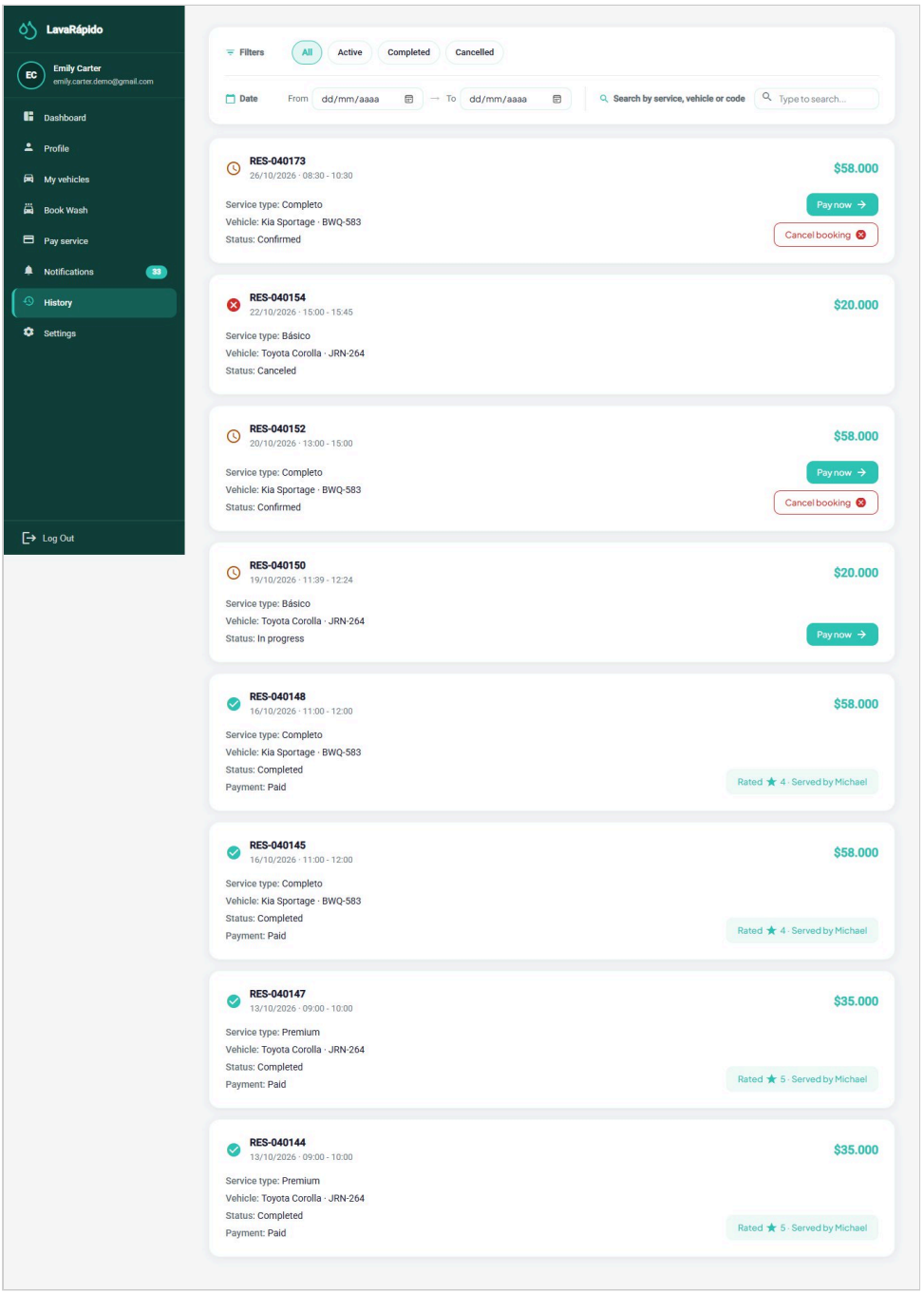
Table 17 *Payment Requirements (payment-service)*

ID	Requirement	Acceptance criteria	Priority	Status
RF-034	**Site payment accounts.** The system must allow the administrator to create, edit, activate and deactivate the payment accounts for each digital method (holder, account number, QR image and instructions).	• Only active accounts are shown to the customer.	Essential	Implemented
RF-013	**Payment methods.** The system must show the customer the site's payment methods (cash, Nequi, Daviplata and bank transfer) and, for each digital method, its QR code and instructions.	• Cash does not show a QR code.	Essential	Implemented
RF-013b	**Receipt upload.** For a digital method, the system must allow the customer to upload the receipt with the declared amount and the transaction reference code.	• JPG or PNG images of up to 10 MB are accepted; the app compresses them before sending. • The payment goes to Under review and is linked to the booking. • Cash does not require a receipt.	Essential	Implemented
RF-014	**Payment review.** The system must allow the administrator to approve or reject a payment under review by comparing the declared amount with the amount due, and to record cash payments directly.	• Rejection requires a reason; the customer is notified and can send another receipt. • A booking can have at most one approved payment. • Approval credits the loyalty points (RF-026).	Essential	Implemented
RF-015	**Payment history.** The system must show the customer their payment history sorted by date and linked to each service.	• Payments are sorted from the most recent booking to the oldest. • They can be filtered by booking date range and by service name (`GET /payments/me?from=&to=&service=`); a start date later than the end date is rejected with `INVALID_DATE_RANGE`. • With no payments, a message says so.	Desirable	Implemented
RF-016	**Payment constraints.** The system must not capture or store card data and must reject a receipt whose reference code already appears in the payment of another booking.	• The rejection returns the error code `TRANSACTION_REFERENCE_REUSED`.	Essential	Implemented
RF-035	**Refund.** The system must allow the administrator to mark an approved payment as refunded.	• Only an Approved payment can move to Refunded; the `payment.refunded` event is published.	Optional	Implemented

Evidence:

RF	Screen
RF-013, RF-013b, RF-016, RF-025 (payment, coupon)	
RF-014, RF-035 (review and refund)	

RF	Screen
RF-034 (site payment accounts)	

RF	Screen
RF-015 (history, within bookings)	

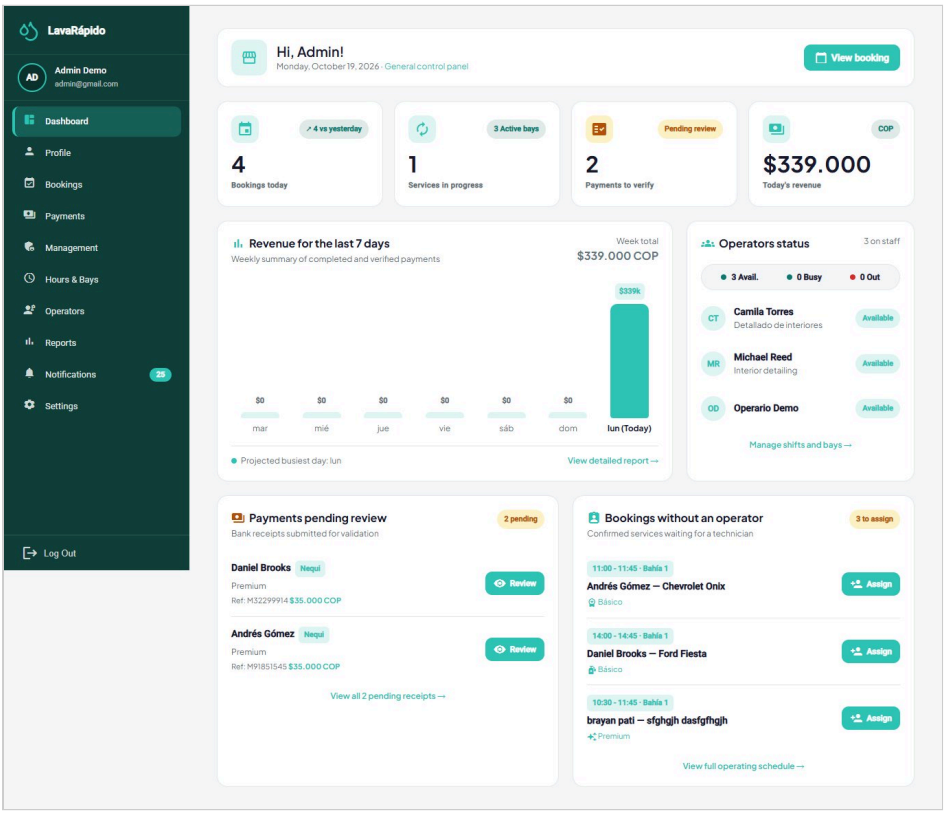
5.8 Dashboard, Notifications, Preferences and Reports

Table 18 *Dashboard, Notification, Preference and Report Requirements*

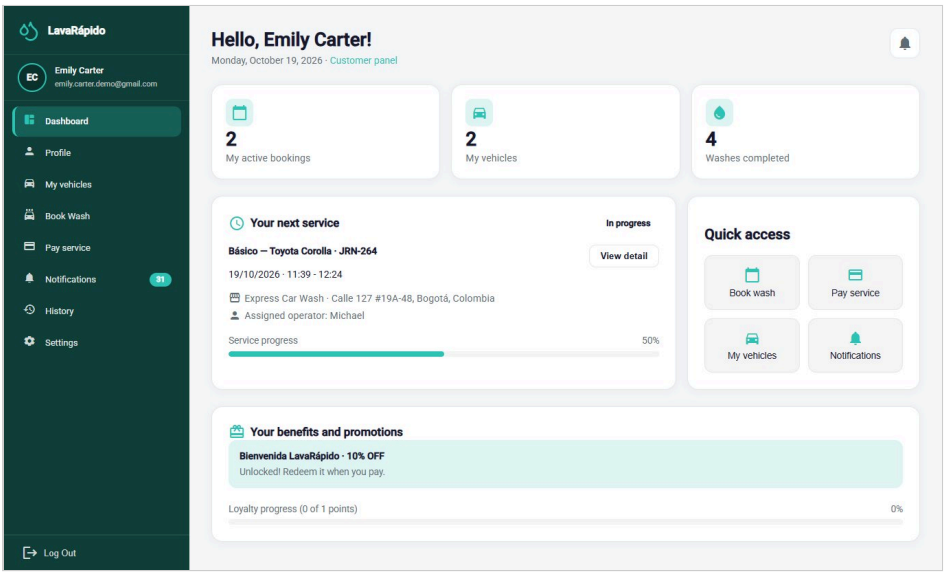
ID	Requirement	Acceptance criteria	Priority	Status
RF-017	**Dashboard per role.** The system must show each role a dashboard with summary metrics and shortcuts.	• With no data, a message says so instead of an empty screen. • The administrator dashboard compares today's bookings with the previous day's, counted by booking-service in the car wash's time zone. • The customer dashboard shows their next service with the name of the assigned operator.	Desirable	Implemented
RF-018	**Notifications.** The system must notify users about relevant events in an inbox with read/unread status, by push in the mobile app and by email for booking events.	• Types: welcome, booking created, confirmed and cancelled, reminder, operator assigned, service started and finished, payment confirmed, rejected and refunded, points earned, promotion available and redeemed, inspection report shared, new app version notice, and system message; staff also get notices of a service cancelled, rescheduled or taken away (operator) and of a booking created, rescheduled, cancelled or without an operator (administrator). • Reminders are sent 24 hours and 1 hour before the booking and are cancelled when the service starts or finishes. • The administrators get an alert when a booking starts in less than 2 hours and still has no operator. • Each notification is written in the language of the account that receives it. • Opening a notification marks it as read.	Desirable	Implemented
RF-019	**Visual theme per account.** The system must allow choosing between four color palettes and keep the choice in the user's account.	• When another account signs in on the same device, that account's theme is applied.	Optional	Implemented
RF-020	**Language per account.** The system must allow switching the interface language between Spanish, English, French and Portuguese, and keep it in the account.	• Without a session, the chosen language is remembered in the browser.	Optional	Implemented

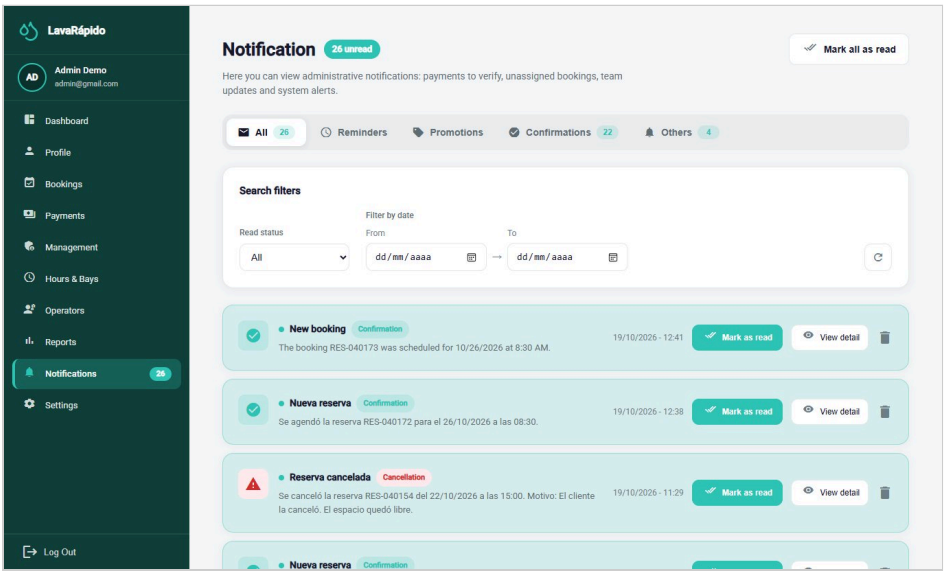
ID	Requirement	Acceptance criteria	Priority	Status
RF-023	**Operational reports.** The system must generate reports on services performed, revenue, average rating and performance per operator for a chosen period.	• With no data in the period, a message says so.	Desirable	**Partial.** The web reads real bookings and payments, and operations-service computes each operator's weekly metrics (services, punctuality, revenue and worked hours) and the team summary; consolidated reports for a chosen period are missing.

Evidence:

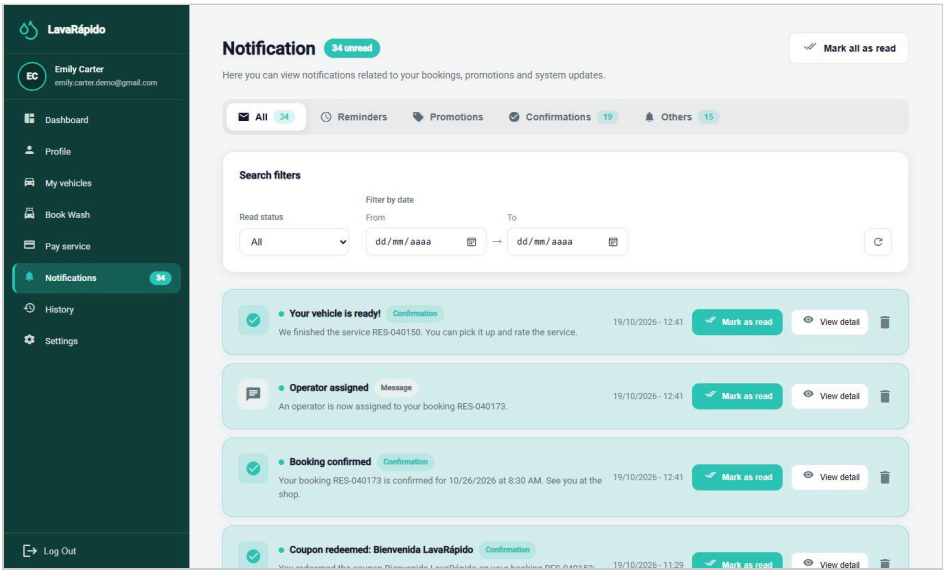
RF	Screen
RF-017 (admin / customer / operator dashboard)	

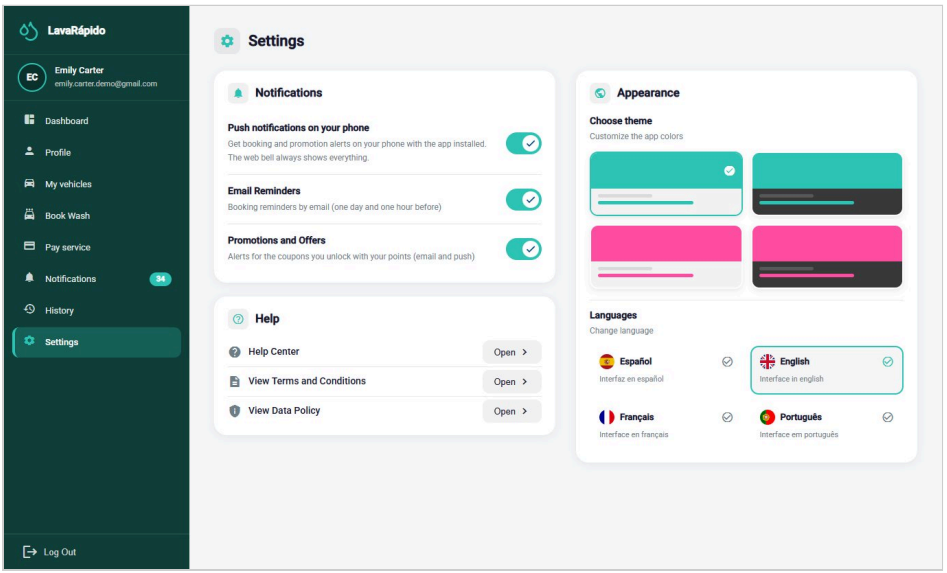
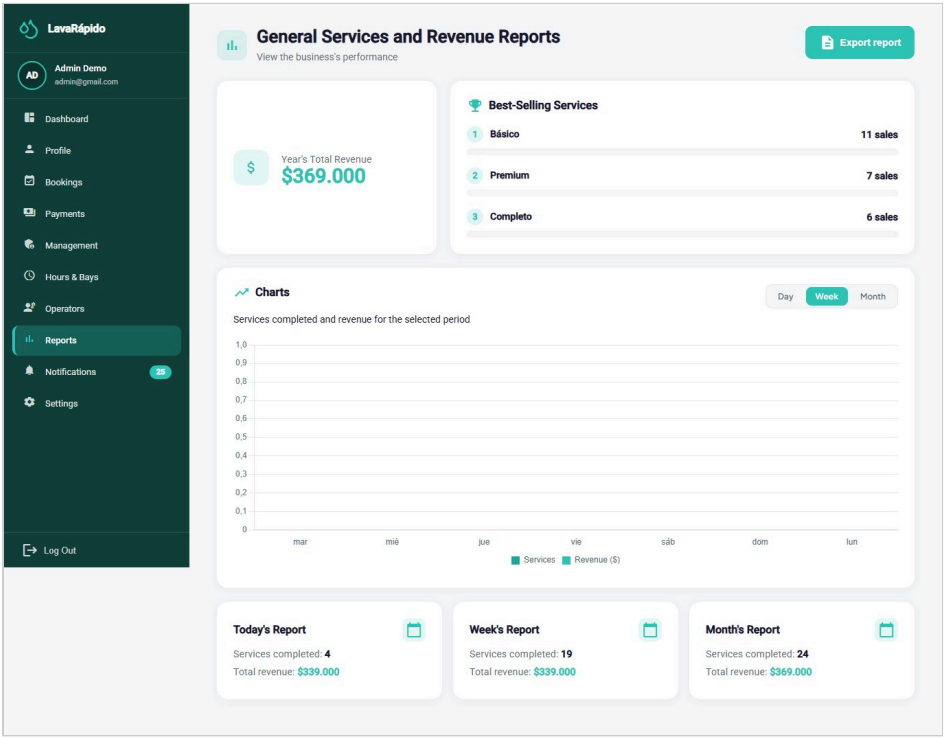
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RF	Screen
RF-018 (notifications)	

.



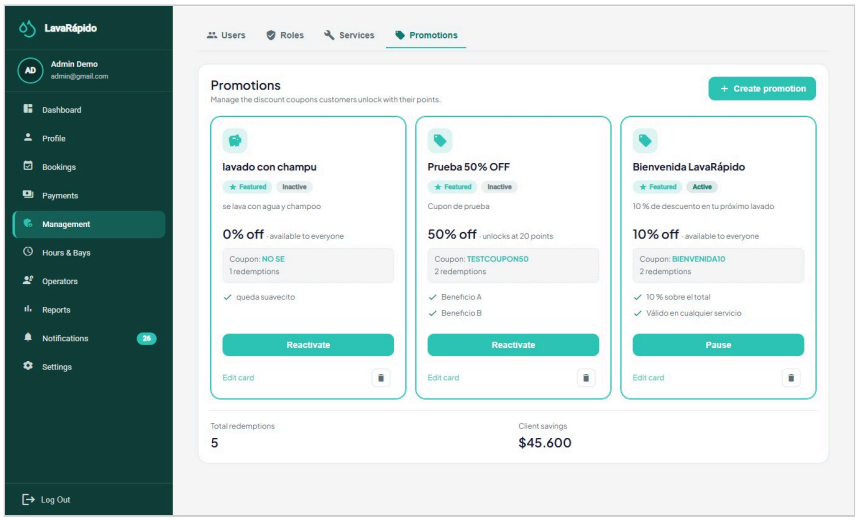
RF	Screen
RF-019, RF-020 (theme and language)	
RF-023 (reports)	

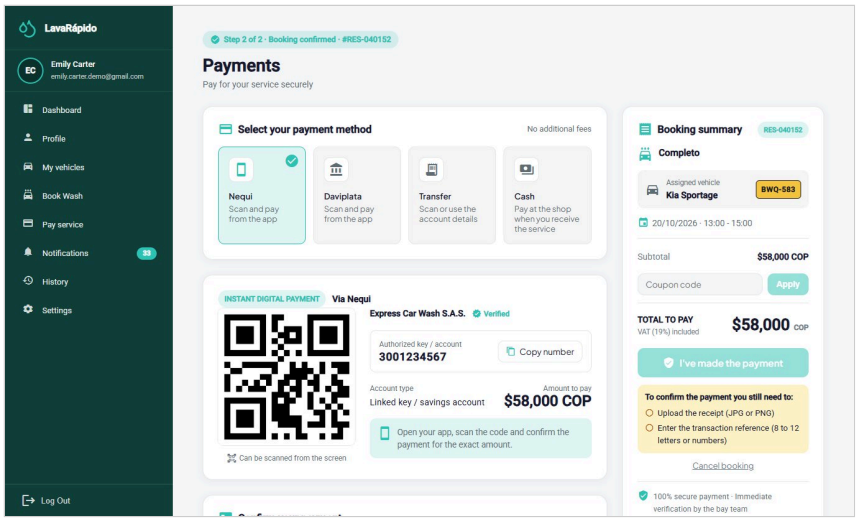
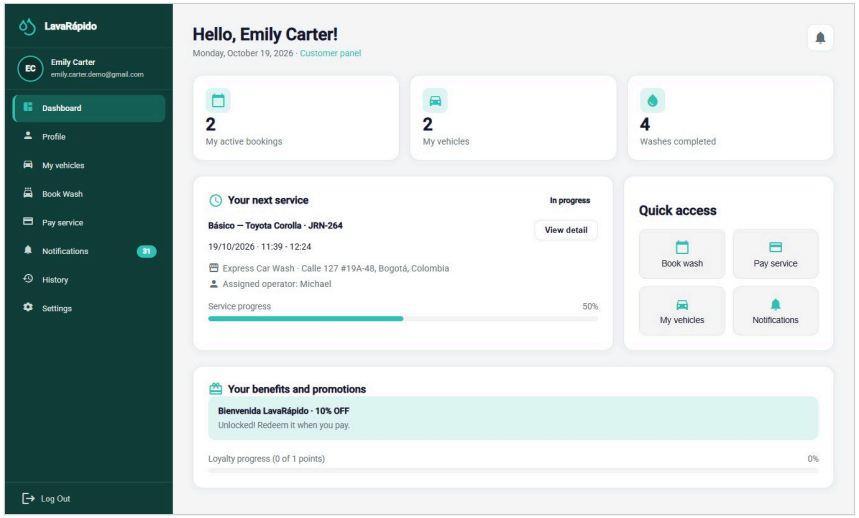
5.9 Promotions and Loyalty

Table 19 *Promotion and Loyalty Requirements (payment-service)*

ID	Requirement	Acceptance criteria	Priority	Status
RF-024	**Promotion management.** The system must allow the administrator to create, edit, activate, deactivate and delete promotions with a code, validity period, discount percentage (1 to 100) and required points.	• A deactivated or expired promotion cannot be applied to new payments.	Desirable	Implemented
RF-025	**Applying a promotion at payment.** The system must allow the customer to apply the code of a valid promotion when paying for a booking.	• The system checks that the promotion is active, within its validity period and unlocked by the customer's points. • The discount is calculated on the booking total and never exceeds it. • If the promotion does not apply, the customer is told and the original amount is kept; if it applies, the redemption is notified in the app and by email.	Optional	Implemented
RF-026	**Loyalty points.** The system must credit the customer with the points of each paid service and use the accumulated balance to unlock promotions.	• Points are credited only once per booking, when its payment is approved. • The balance is calculated from the ledger of movements and never from a copy in the profile. • Redeeming a promotion does not consume points.	Optional	Implemented

Evidence:

RF	Screen
RF-024 (promotion management)	

RF	Screen
RF-025, RF-026 (redeeming a coupon / points balance on the customer dashboard)	 . 

5.10 Business Rules

Business rules are conditions that the system must always keep, regardless of the screen or the service that performs the operation. Table 20 lists them with their identifier from the domain model.

Table 20 *System Business Rules*

ID	Rule	Requirement
INV-BOOK-001/002	A booking requires a vehicle that belongs to the customer making the booking.	RF-005

ID	Rule	Requirement
INV-BOOK-003	A booking has at least one service line; the same rate is not repeated within a booking.	RF-005
INV-BOOK-004	The booking end time is later than the start time.	RF-005
INV-BOOK-005/006	A booking is only confirmed within opening hours and with an active bay without overlapping bookings.	RF-006
INV-BOOK-007	Only Scheduled or Confirmed bookings can be rescheduled or cancelled; the customer can only cancel before the start time.	RF-007
INV-BOOK-008	Each line keeps the price in force when the booking was made, even if the catalog changes later.	RF-031
INV-EXEC-001/002	Each service line has at most one execution and one responsible operator.	RF-009
INV-EXEC-003	An operator cannot have overlapping services.	RF-009
INV-EXEC-004	Only the assigned operator updates the progress of their service.	RF-012b
RN-EXEC-01	A service can only be started from 15 minutes before the booking time.	RF-012b
RN-ASSIGN-01	Automatic assignment respects the customer's choice if that operator is free; otherwise it picks the one with the fewest bookings that day and, in a tie, the one with the lowest identifier.	RF-008, RF-010
RN-AUTH-01	Five wrong passwords within 15 minutes lock the account for 15 minutes.	RF-004b
RN-INSP-01	The inspection report only applies to bookings of 180 minutes or more and can only be edited while the vehicle is at the car wash (booking Confirmed or In progress).	RF-036
INV-RATING-001 to 004	The rating is unique, a whole number from 1 to 5, and given after completion; the comment requires a score.	RF-012
INV-PAY-001	Every payment belongs to a booking; a booking has at most one approved payment.	RF-014
INV-PAY-002/003	Digital payments require a receipt; cash payments are recorded by the administrator.	RF-013b
INV-PAY-005	A difference between the declared amount and the amount due is a reason for rejection, never for automatic approval.	RF-014
INV-PAY-006	A reference code is not reused across bookings.	RF-016
INV-PAY-007	The system does not capture or store card data.	RF-016
INV-LOY-001	The points balance is always calculated from the ledger of movements.	RF-026
RN-PROMO-01	The discount is a percentage (1 to 100), is applied to the booking total and never exceeds it.	RF-025

6. Non-Functional Requirements

Non-functional requirements are classified according to the product quality characteristics of ISO/IEC 25010 (ISO/IEC, 2023). Each one has a metric or criterion that can be verified in the

local Docker environment (see Table 21). The requirements of 1,000 concurrent users in the cloud and 99% availability from earlier versions were withdrawn, because the system is only published on a server for the academic review and does not serve the real operation of a car wash.

Table 21 *Non-Functional Requirements*

ID	Characteristic	Requirement and metric	Verification	Status
RNF-001	Performance efficiency	The 95th percentile of the response time of the booking and authentication endpoints is under 2 seconds, and the availability query responds in under 2 seconds with the test data set.	Script `lavarapido-infra/e2e/performance.js`: 30 requests per endpoint through the gateway in the local environment	**Implemented.** Measured on October 14, 2026: the highest 95th percentile was 686 ms (`/payments/me`).
RNF-002	Availability	All services answer UP on their health endpoint (`/actuator/health` or `/health`) during every academic review session.	Check before each session	**Implemented.** Verified live on October 9, 2026: the 6 services and the gateway answered `200 OK`.
RNF-003	Integrity and concurrency	Under simultaneous requests, only one booking takes a bay in a slot; a busy operator is not assigned; a booking does not have two approved payments (filtered unique index) and a reference code is not repeated.	Integration tests	Partial: rules implemented; integration tests pending
RNF-004	Security	Private endpoints with JWT (401 without a token or with an expired token); token validity of 1 hour; BCrypt passwords of 8 to 72 characters, with at least one uppercase letter, one number and one special character, and no spaces; access by role (403 between roles) and by resource owner; lockout after 5 failed attempts; secrets only in environment variables; HTTPS outside development.	Web tests by role, password policy and lockout tests, a test that the gateway does not expose `/internal`, code review	**Implemented**; on the public server Caddy serves everything over HTTPS.
RNF-005	Maintainability: observability	Each service writes requests and errors to standard output with the `X-Correlation-Id` correlation identifier.	Log review with `docker logs`	Partial

ID	Characteristic	Requirement and metric	Verification	Status
RNF-006	Maintainability	Every change goes through a Pull Request with at least one approval, uses Conventional Commits and versions schema changes with Liquibase; frontend and backend communicate only through the API.	GitHub history and code review	Partial
RNF-007	Portability	A new team member brings up the whole solution in Docker containers by following the local installation guide, on any of the team's operating systems.	Installation from scratch on more than one computer	Partial
RNF-008	Compatibility	The app works on Android 8.0 or later; the web works on the current versions of Chrome, Firefox and Edge, and is usable from 360 px wide.	Script `lavarapido-infra/e2e/width-360.js` for the web; manual tests on devices and emulators	Partial: on October 17, 2026, none of the 60 web screens measured (360 and 768 px) or of the 26 mobile app screens (360 px, in their web version) scrolls sideways or clips content; testing on an Android 8 device is still missing.
RNF-009	Usability	A customer completes a booking in 5 steps or fewer from the dashboard; each form points out the invalid field; each action that changes data shows a confirmation or an error.	Review of flows and forms	Partial: booking takes 5 steps (vehicle, service, date, time and confirm; choosing the operator is optional) and the form messages were reviewed in the 4 languages; a test with users is still missing.
RNF-010	Usability: localization	All screens are available in the four languages with no untranslated keys; amounts in COP; dates in the format of each language.	Manual review per language	Partial
RNF-011	Reliability: idempotency	A domain event received more than once produces a single effect (one notification, one points credit).	Unit tests of the event consumer and the points credit	Implemented

ID	Characteristic	Requirement and metric	Verification	Status
RNF-012	Reliability: fault tolerance	A failure of the SMTP server, Expo Push or RabbitMQ does not prevent confirming a booking or approving a payment; the notification stays in the inbox.	Test with the external service stopped	Implemented

7. Use Case Model

7.1 Use Case Diagram

The 32 use cases are presented in four diagrams, one per actor, grouped by module (Figures 3a to 3d).

Figure 3a *Use Cases Common to the Three Roles*

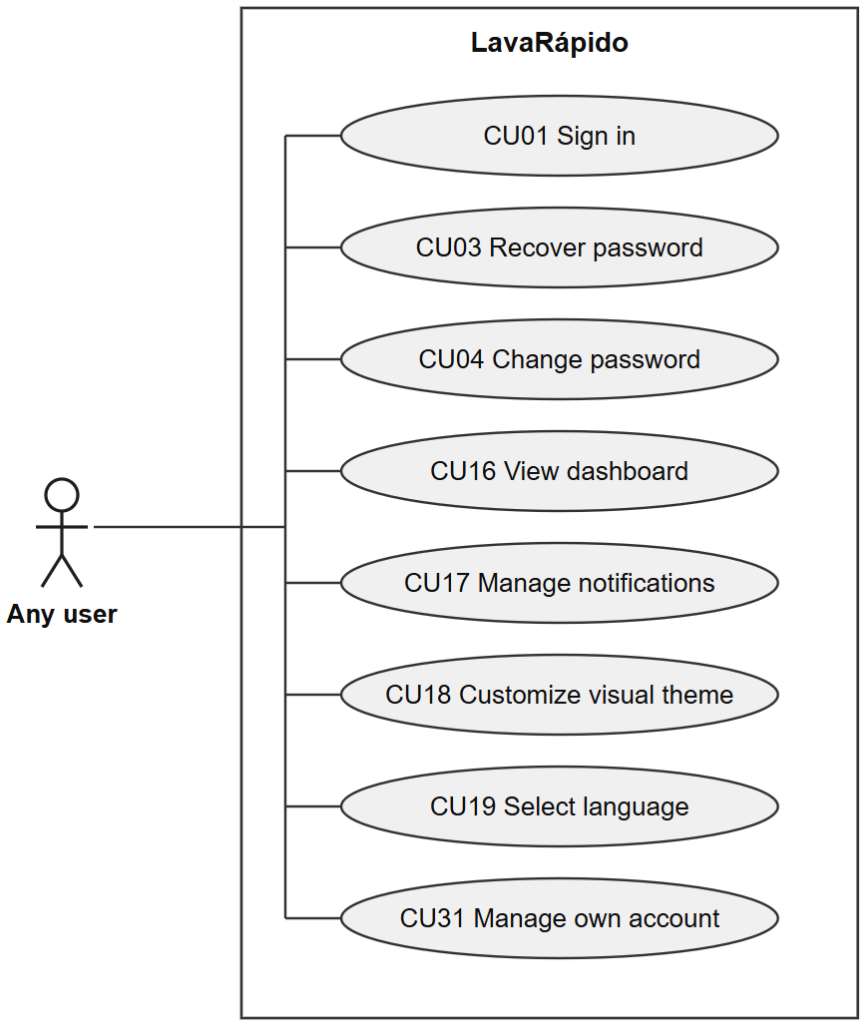
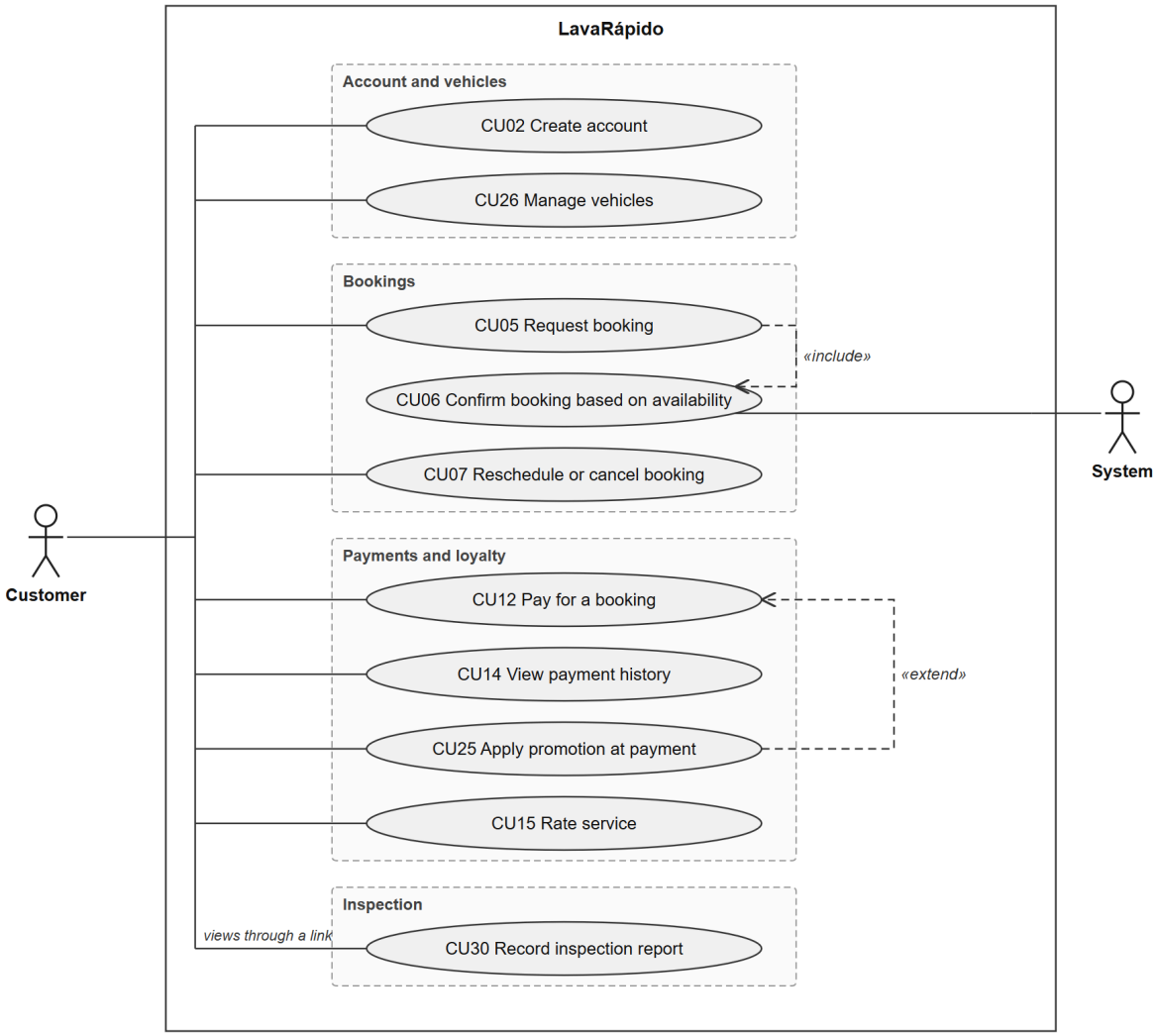


Figure 3b *Customer Use Cases*



Note. «include» means that CU05 always runs CU06; «extend» means that CU25 optionally extends CU12. The customer does not record the inspection report (CU30): they view it through the link the administrator shares.

Figure 3c *Operator Use Cases*

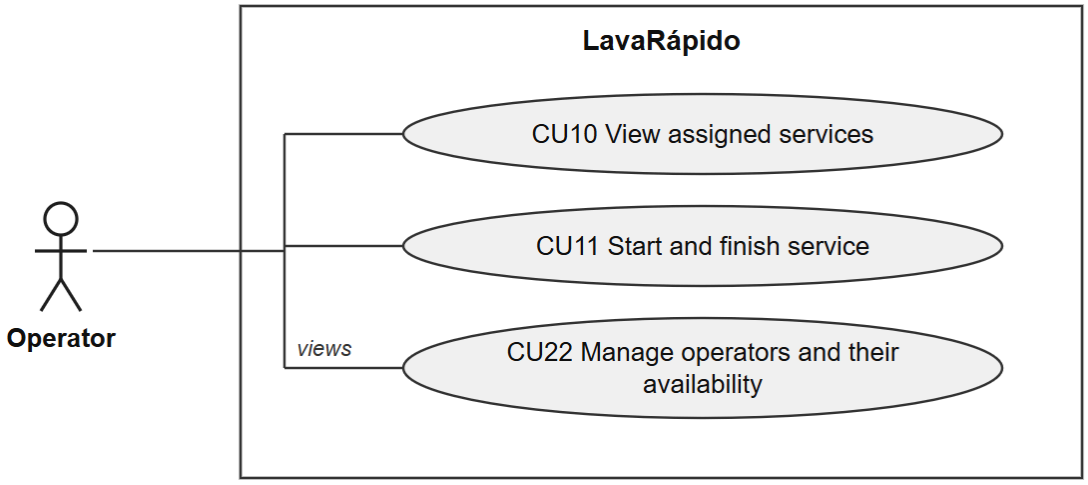
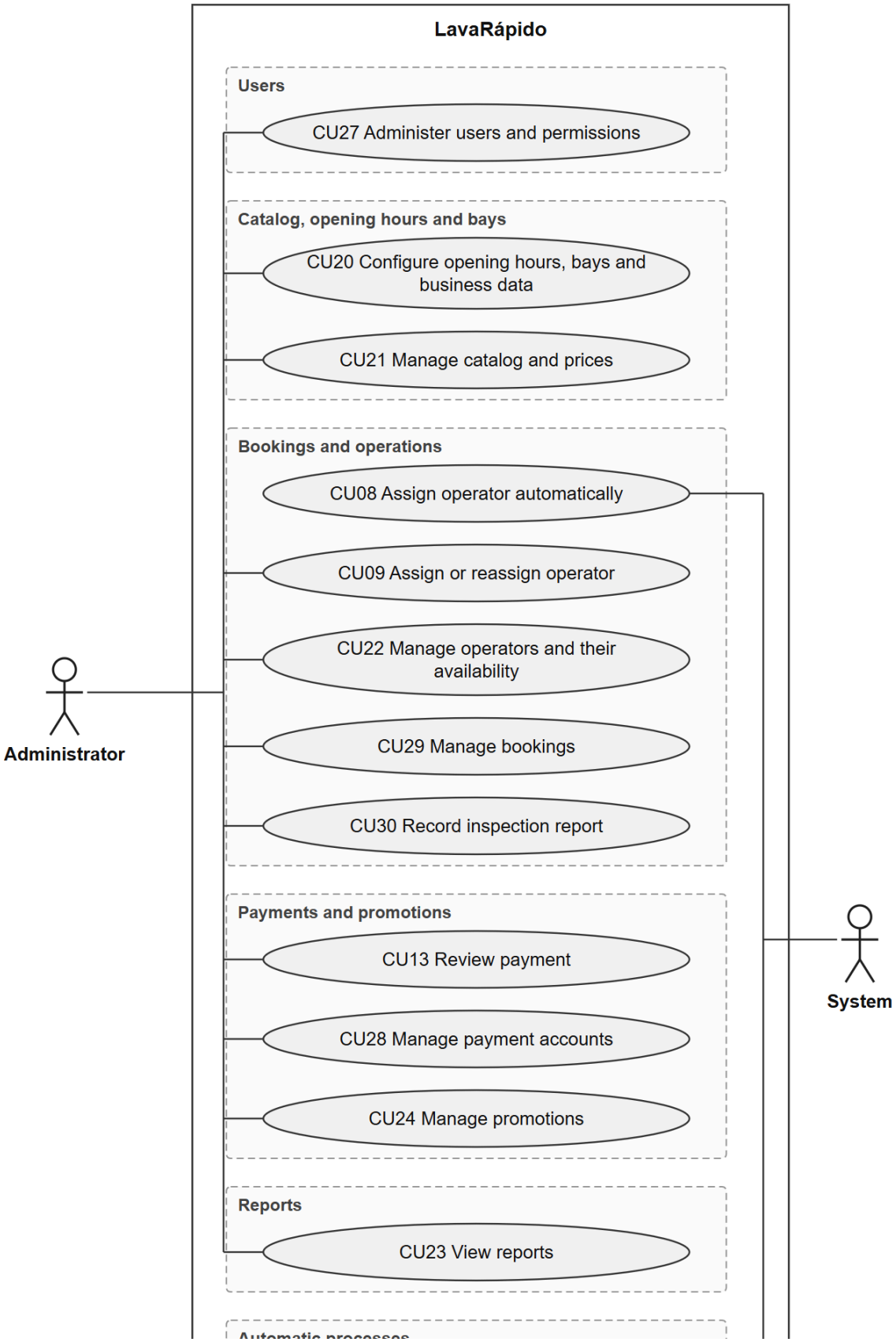
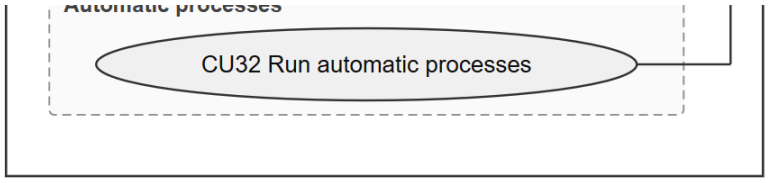


Figure 3d *Administrator and System Use Cases*


LavaRápido
Users
CU27 Administer users and permissions
Catalog, opening hours and bays
CU20 Configure opening hours, bays and business data
CU21 Manage catalog and prices
Bookings and operations
CU08 Assign operator automatically
CU09 Assign or reassign operator
CU22 Manage operators and their availability
CU29 Manage bookings
CU30 Record inspection report
Payments and promotions
CU13 Review payment
CU28 Manage payment accounts
CU24 Manage promotions
Reports
CU23 View reports
Administrator
System



Note. CU22 appears as a view-only use case in the operator diagram and as full management in the administrator diagram, since the administrator is the one who records shifts, absences and the operator profile (RF-022, RF-038). CU08 is run by the system each time a booking is confirmed, and CU32 groups the processes that run without anyone asking for them (reminders, alerts, notices and the first administrator). Diagrams by the authors.

7.2 Actor and Use Case Matrix

Table 22 *Use Cases by Actor*

Actor	Use cases
All roles	CU01, CU03, CU04, CU16, CU17, CU18, CU19, CU31
Customer	CU02, CU05, CU06, CU07, CU12, CU14, CU15, CU25, CU26, CU30 (view only)
Operator	CU10, CU11, CU22 (view only)
Administrator	CU09, CU13, CU20, CU21, CU22, CU23, CU24, CU27, CU28, CU29, CU30
System	CU06 (included in CU05), CU08, CU32

7.3 Use Case Specifications

Each use case is specified with the Cockburn (2001) format: actors, precondition, main flow, alternative flows, postcondition, requirements covered and status.

Table 23 *CU01: Sign in*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user has an active account.
Main flow	1. The user enters email and password. 2. The system validates the credentials and issues the token. 3. The system redirects to the role's dashboard.
Alternative flows	• Invalid credentials: a generic message is shown. • Five failed attempts in 15 minutes: the account is locked for 15 minutes and the system offers password recovery, which lifts the lockout.
Postcondition	The user stays authenticated while the token is valid (1 hour).
Requirements	RF-002, RF-004b
Status	Implemented

Table 24 *CU02: Create account*

Field	Description
Actors	Customer.
Precondition	There is no account with the same email.
Main flow	1. The customer fills in their personal data and password. 2. They accept the data processing policy. 3. The system creates the account and the profile, and sends the welcome email.
Alternative flows	• Email already registered: the customer is informed and the account is not created. • Password that does not meet the policy: the broken rules are shown.
Postcondition	The account and the customer profile are created.
Requirements	RF-001
Status	Implemented

Table 25 *CU03: Recover password*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user has access to their account's email.
Main flow	1. The user enters their email. 2. The system sends a 6-digit code. 3. The user types the code and the new password.
Alternative flows	• Expired code or 5 failed attempts: the user requests a new code. • Unregistered email: the response is identical and no email is sent. • New password equal to the current one: it is rejected and the user can try another one without requesting a new code.
Postcondition	The password is updated, the code is no longer valid and, if the account was locked, the lockout is lifted.
Requirements	RF-003
Status	Implemented

Table 26 *CU04: Change password*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user is authenticated.
Main flow	1. The user types the current password and the new one. 2. The system validates and saves the change.
Alternative flows	• Wrong current password or invalid new password: the change is rejected.
Postcondition	The password is updated.
Requirements	RF-004
Status	Implemented

Table 27 *CU05: Request booking*

Field	Description
Actors	Customer.
Precondition	The customer is authenticated and has at least one registered vehicle.
Main flow	1. The customer chooses the vehicle. 2. They choose one or more services; the system shows the price according to the vehicle type. 3. They choose date and time. 4. If they want, they choose an operator from the list of those free at that time. 5. The system runs CU06.
Alternative flows	• No registered vehicles: the system asks the customer to register one (CU26). • No one free at that time: the booking goes ahead without an operator preference.
Postcondition	The request is ready for the availability check.
Requirements	RF-005, RF-010
Status	Implemented

Table 28 *CU06: Confirm booking based on availability*

Field	Description
Actors	System (included in CU05).
Precondition	There is a booking request in progress.
Main flow	1. The system checks the opening hours, the exceptions and the active bays. 2. With a free bay, it confirms the booking and assigns the bay. 3. It publishes the confirmation: the customer is notified, the reminders are scheduled and CU08 runs.
Alternative flows	• No free bay: nothing is saved and up to 5 alternative times on the same day are offered.
Postcondition	The booking is Confirmed and the bay is taken in that slot.
Requirements	RF-006
Status	Implemented

Table 29 *CU07: Reschedule or cancel booking*

Field	Description
Actors	Customer.
Precondition	The booking is in the Scheduled or Confirmed status.
Main flow	1. The customer opens the booking from their history. 2. They choose to cancel it and confirm in the dialog (no reason is written). 3. The system cancels it with the reason `CUSTOMER_REQUEST`, frees the bay and sends a notification.
Alternative flows	• Service in progress or finished, or the start time has passed: the cancellation is blocked with an explanatory message. • To change the date or time, the customer cancels and books again, or asks the car wash: rescheduling is done by the administrator (CU29).
Postcondition	The booking is cancelled and the bay is freed.
Requirements	RF-007
Status	Implemented

Table 30 *CU08: Assign operator automatically*

Field	Description
Actors	System.
Precondition	The booking is Confirmed and has no operator.
Main flow	1. When the booking confirmation arrives, the system identifies the eligible operators for the slot. 2. If the customer chose an operator who is free, it assigns them; otherwise it picks the one with the fewest bookings that day (tie: lowest identifier). 3. It notifies the customer and the operator.
Alternative flows	• No eligible operators: the booking stays without an operator and, when less than 2 hours remain, the administrators get an alert so they can run CU09. • Rescheduled booking that already had an operator: the operator is kept.
Postcondition	The service has an assigned and notified operator.
Requirements	RF-008, RF-010
Status	Implemented

Table 31 *CU09: Assign or reassign operator*

Field	Description
Actors	Administrator.
Precondition	The booking has not started.
Main flow	1. The administrator opens the booking. 2. The system shows the candidates: available ones first, by name, with their rating. 3. The administrator picks an operator and confirms. 4. The system notifies the operator.
Alternative flows	• Operator not available: the assignment is blocked and the reason is shown. • Execution already started: reassignment is blocked.
Postcondition	The booking is assigned to the chosen operator.
Requirements	RF-009
Status	Implemented

Table 32 *CU10: View assigned services*

Field	Description
Actors	Operator.
Precondition	The operator is authenticated.
Main flow	1. The operator opens their dashboard. 2. They see their bookings with customer, vehicle, time, services and bay.
Alternative flows	• No assigned services: a message says so.
Postcondition	The operator knows their work.
Requirements	RF-011
Status	Implemented

Table 33 *CU11: Start and finish service*

Field	Description
Actors	Operator.
Precondition	The booking is assigned to the operator, is Confirmed and 15 minutes or less remain before its start time.
Main flow	1. The operator marks the service as started. 2. When done, they mark it as finished.
Alternative flows	• Booking of another operator, or before the 15-minute window: the action is blocked.
Postcondition	The booking is Completed, the pending reminders are cancelled and the customer can rate it.
Requirements	RF-012b
Status	Implemented

Table 34 *CU12: Pay for a booking*

Field	Description
Actors	Customer.
Precondition	The booking is confirmed and has no approved payment.
Main flow	1. The customer chooses the payment method. 2. For a digital method, they see the QR code and the instructions. 3. They upload the receipt with the declared amount and the reference code. 4. The payment goes to Under review.
Alternative flows	• Cash: no receipt is uploaded and the administrator records the payment. • Reference code already used in another booking: the receipt is rejected. • The customer can apply a promotion (CU25).
Postcondition	The payment is recorded and linked to the booking.
Requirements	RF-013, RF-013b, RF-016
Status	Implemented

Table 35 *CU13: Review payment*

Field	Description
Actors	Administrator.
Precondition	There is a payment under review or a cash payment to record.
Main flow	1. The administrator opens the pending payments. 2. They compare the declared amount with the amount due. 3. They approve or reject the payment.
Alternative flows	• Different amount or unreadable receipt: they reject it with a reason and the customer can resend. • Approved payment that must be returned: they mark it as refunded (RF-035).
Postcondition	The payment is approved or rejected and the customer is notified; approval credits points.
Requirements	RF-014, RF-035
Status	Implemented

Table 36 *CU14: View payment history*

Field	Description
Actors	Customer.
Precondition	The customer is authenticated.
Main flow	1. The customer opens the history. 2. They see their payments with date, service, amount and status, from the most recent booking to the oldest. 3. If they want, they filter by dates or by service.
Alternative flows	• No payments: a message says so.
Postcondition	The customer knows the status of their payments.
Requirements	RF-015
Status	Implemented

Table 37 *CU15: Rate service*

Field	Description
Actors	Customer.
Precondition	The booking is Completed and has not been rated.
Main flow	1. The customer picks a score from 1 to 5. 2. They write an optional comment. 3. They save the rating.
Alternative flows	• Booking not completed or already rated: the system rejects the rating.
Postcondition	The rating is linked to the booking and its lines.
Requirements	RF-012
Status	Implemented

Table 38 *CU16: View dashboard*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user is authenticated.
Main flow	1. The user enters the system. 2. They see the summary and the shortcuts of their role.
Alternative flows	• No data: a message says so.
Postcondition	The user has an overview of their activity.
Requirements	RF-017
Status	Implemented

Table 39 *CU17: Manage notifications*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user is authenticated.
Main flow	1. The user opens the inbox, sorted from newest to oldest. 2. They see the notifications with their status. 3. Clicking one opens its detail and marks it as read. 4. If the notification refers to a section (for example, a booking or a payment), the "Go to…" button takes them to that screen.
Alternative flows	• No notifications: the inbox shows a message saying so.
Postcondition	The user is informed about the relevant events.
Requirements	RF-018
Status	Implemented

Table 40 *CU18: Customize visual theme*

Field	Description
Actors	Customer, operator, administrator.
Precondition	The user is authenticated.
Main flow	1. The user picks one of the four palettes. 2. The system applies it and saves it in the account.
Alternative flows	• Saving fails: the theme is applied on the device and the account keeps its previous one.
Postcondition	The theme is linked to the account.
Requirements	RF-019
Status	Implemented

Table 41 *CU19: Select language*

Field	Description
Actors	Any user, with or without a session.
Precondition	None.
Main flow	1. The user picks the language. 2. The system updates the interface texts.
Alternative flows	• Without a session: the language is only remembered in the browser.
Postcondition	The interface is shown in the chosen language; with a session, it is saved in the account.
Requirements	RF-020
Status	Implemented

Table 42 *CU20: Configure opening hours, bays and business data*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.

Field	Description
Main flow	1. They set the weekly opening hours and the daily break. 2. They record exceptions (public holidays, special hours). 3. They create, edit or deactivate bays. 4. They update the legal, tax and service data of the business.
Alternative flows	• Closing time earlier than or equal to the opening time: the change is rejected. • Missing legal name, tax ID or address, or an invalid email or phone: the business data is not saved.
Postcondition	The opening hours and the bays apply to later bookings; the business data is shown on the web and in the app.
Requirements	RF-021, RF-032, RF-037
Status	Implemented

Table 43 *CU21: Manage catalog and prices*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.
Main flow	1. They create or edit a service. 2. They set the price and duration by vehicle type, and its points. 3. They activate or deactivate the service.
Alternative flows	• Required data is missing or the rate overlaps another active one: the system says so before saving.
Postcondition	The catalog is updated; existing bookings keep their price.
Requirements	RF-031
Status	Implemented

Table 44 *CU22: Manage operators and their availability*

Field	Description
Actors	Administrator (operator: view only).
Precondition	The user is authenticated.
Main flow	1. The administrator records the operator's weekly shift. 2. They record or delete absences. 3. They edit the operator's specialty and skills. 4. They activate or deactivate the operator. 5. They check the operator's weekly metrics and the team summary.
Alternative flows	• Absence that overlaps an assigned service: the system warns about it.
Postcondition	The availability is updated for automatic and manual assignment.
Requirements	RF-022, RF-038
Status	Implemented

Table 45 *CU23: View reports*

Field	Description
Actors	Administrator.

Field	Description
Precondition	The administrator is authenticated.
Main flow	1. They open the reports section. 2. They choose the type of report and the period.
Alternative flows	• No data in the period: a message says so.
Postcondition	The administrator has information to make decisions.
Requirements	RF-023
Status	Partial

Table 46 *CU24: Manage promotions*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.
Main flow	1. They create or edit a promotion with a code, validity period, percentage and required points. 2. They activate or deactivate it.
Alternative flows	• Data is missing or the percentage is outside 1 to 100: the system says so before saving.
Postcondition	The promotion becomes available or disabled.
Requirements	RF-024
Status	Implemented

Table 47 *CU25: Apply promotion at payment*

Field	Description
Actors	Customer (extends CU12).
Precondition	The customer has a booking to pay and an unlocked promotion.
Main flow	1. During payment, the customer enters the promotion code. 2. The system checks validity, status and points, and calculates the discount. 3. The customer pays the discounted amount.
Alternative flows	• Expired or inactive promotion, or not enough points: the customer is told and the original amount is kept.
Postcondition	The redemption is recorded in the booking and the customer is notified.
Requirements	RF-025, RF-026
Status	Implemented

Table 48 *CU26: Manage vehicles*

Field	Description
Actors	Customer.
Precondition	The customer is authenticated.
Main flow	1. The customer registers a vehicle with license plate and type. 2. They view, edit or remove their vehicles.

Field	Description
Alternative flows	• Incomplete data: the system points out the missing field.
Postcondition	The vehicles are available for booking.
Requirements	RF-030
Status	Implemented

Table 49 *CU27: Administer users and permissions*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.
Main flow	1. They list the users. 2. They create operator or administrator accounts. 3. They activate or deactivate accounts. 4. They edit a role's permission matrix.
Alternative flows	• Email already registered: the account is not created. • They try to deactivate an administrator account or their own: the system prevents it and explains why.
Postcondition	The accounts and the permission matrix are updated.
Requirements	RF-028, RF-029
Status	Implemented

Table 50 *CU28: Manage payment accounts*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.
Main flow	1. They create or edit a payment account with holder, number, QR code and instructions. 2. They activate or deactivate it.
Alternative flows	• Required data is missing: the system asks for it before saving.
Postcondition	Active accounts are shown to the customer at payment.
Requirements	RF-034
Status	Implemented

Table 51 *CU29: Manage bookings*

Field	Description
Actors	Administrator.
Precondition	The administrator is authenticated.

Field	Description
Main flow	1. They view the bookings within a date range. 2. They change the status of a booking (cancel with a reason, mark as "no-show"). 3. If needed, they reschedule it to another date and time, with a new bay and the same prices.
Alternative flows	• Transition not allowed by the life cycle: it is rejected.
Postcondition	The booking is in the new status and the customer is notified.
Requirements	RF-033
Status	Implemented

Table 52 *CU30: Record inspection report*

Field	Description
Actors	Administrator (customer: view only).
Precondition	The booking lasts 180 minutes or more and is Confirmed or In progress.
Main flow	1. The administrator opens the booking's report. 2. They go through the phases in order (reception, exterior, interior, engine and delivery) and mark each one in progress or done. 3. They record the findings with area, note, severity and photos. 4. They publish the report. 5. They share it: the customer gets a notification with the link. 6. The customer opens the link and sees the report with a masked license plate.
Alternative flows	• Booking shorter than 180 minutes, cancelled or pending: it does not accept a report. • Completed booking: the report can be viewed but not edited. • More than 5 photos in a finding: rejected. • Sharing before publishing: rejected.
Postcondition	The report is published and the customer has the link.
Requirements	RF-036
Status	Implemented

Table 53 *CU31: Manage own account*

Field	Description
Actors	Customer, operator or administrator (each with their own account).
Precondition	The user is authenticated.
Main flow	1. They open their profile. 2. They update first and last names, phone and contact address. 3. They change their email by confirming the current password. 4. They sign out. 5. If they will no longer use the account, they deactivate it by confirming the password: all their sessions are closed.
Alternative flows	• Email already registered by another account: it is not changed. • Wrong current password: the email is not changed and the account is not deactivated. • Administrator account: it does not have the option to deactivate itself. • Empty address: it is deleted.
Postcondition	The profile data is updated, or the account is deactivated and keeps its history.
Requirements	RF-027
Status	Implemented

Table 54 *CU32: Run automatic processes*

Field	Description
Actors	System (scheduled tasks and events). The notice of a new app version is started by a team member when they publish the APK.
Precondition	The services and RabbitMQ are running.
Main flow	1. At start-up, if there is no administrator, the system creates the first one with the data of the server configuration. 2. Each booking, operator, payment or inspection event is turned into notifications for the customer, the operator and the administrators, once per event. 3. When a booking is confirmed it schedules its reminders (24 hours and 1 hour before) and sends them on time. 4. Two hours before a booking without an operator it warns the administrators. 5. When a new APK is published it notifies the users who have the app installed.
Alternative flows	• The booking is cancelled, started or finished: the pending reminders are removed. • The service was down and the booking time already passed: the reminder is closed without being sent. • The same app version is published again: the notice is not repeated. • An administrator already exists: no other one is created.
Postcondition	Users get the notices that apply to them, without duplicates.
Requirements	RF-018
Status	Implemented

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8. Business Flows

8.1 Main Flow: From Booking to Rating

The main business flow goes through the system's services in the following order:

1. The customer creates their account (CU02); security-service publishes the registration and customer-service creates the profile; notification-service sends the welcome message.
2. The customer registers their vehicle (CU26) and requests a booking (CU05).
3. booking-service validates opening hours and bays (CU06), confirms the booking and publishes `booking.confirmed`; notification-service sends the notice and schedules the 24-hour and 1-hour reminders.
4. operations-service receives `booking.confirmed` and assigns a free operator (CU08): the one the customer chose if still free or, otherwise, the one with the fewest bookings that day. The customer and the operator are notified. If no one was free, the administrator assigns one by hand (CU09) and gets an alert when less than 2 hours remain.

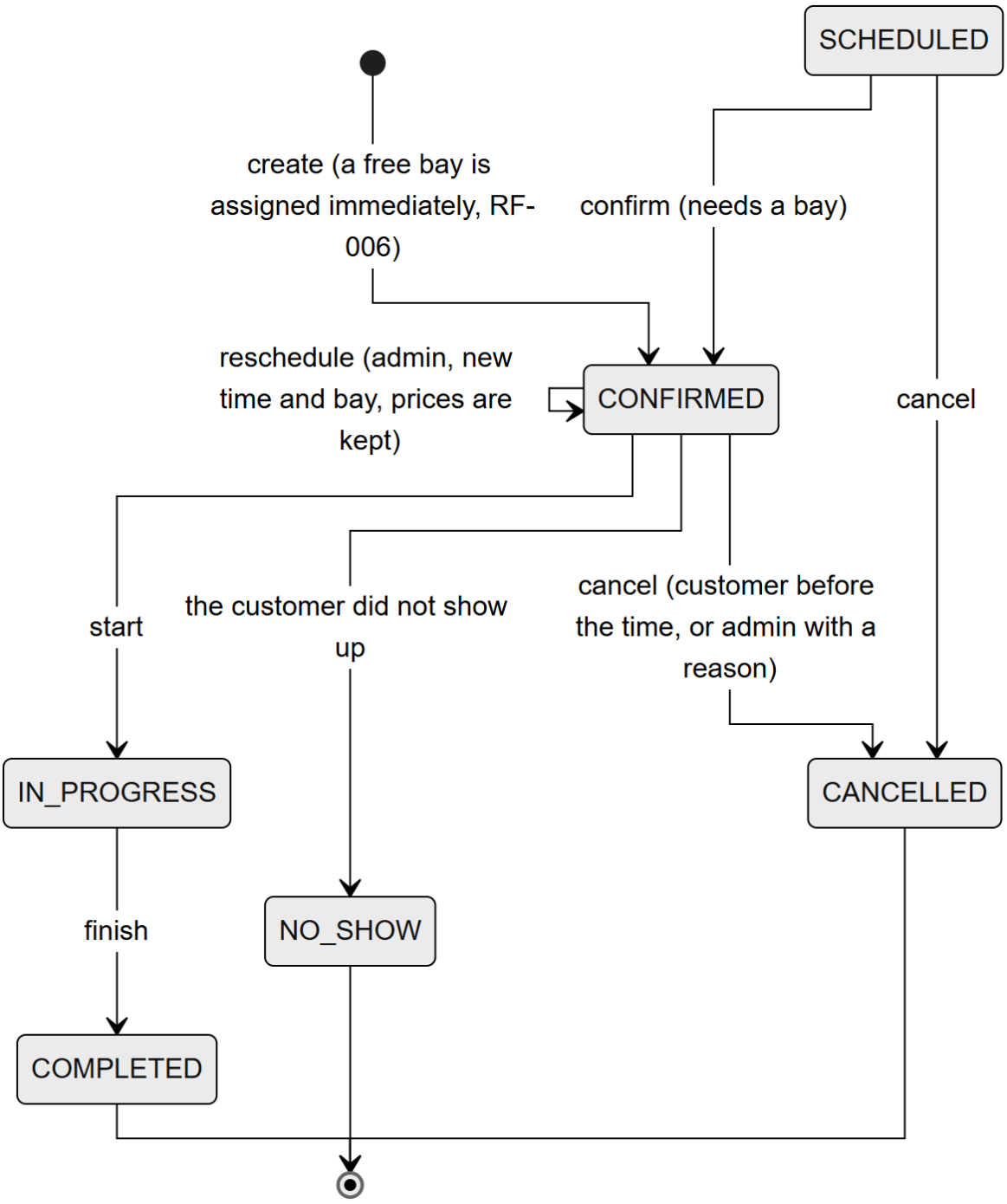
5. The customer pays (CU12): they upload the receipt and, if they want, apply a promotion (CU25). The administrator reviews the payment (CU13); when it is approved, the points are credited and the customer is notified.
6. From 15 minutes before the time, the operator starts the service and finishes it when done (CU11); the booking moves to In progress and then to Completed, and the pending reminders are cancelled. For services of 180 minutes or more, the administrator records the inspection report and shares it with the customer (CU30).
7. The customer rates the service (CU15).

The rule that defines whether payment must be approved before the service starts is still open (section 11.3); today steps 5 and 6 are independent.

8.2 Booking Life Cycle

Figure 4 shows the statuses of a booking and the allowed transitions.

Figure 4 *Booking State Diagram*

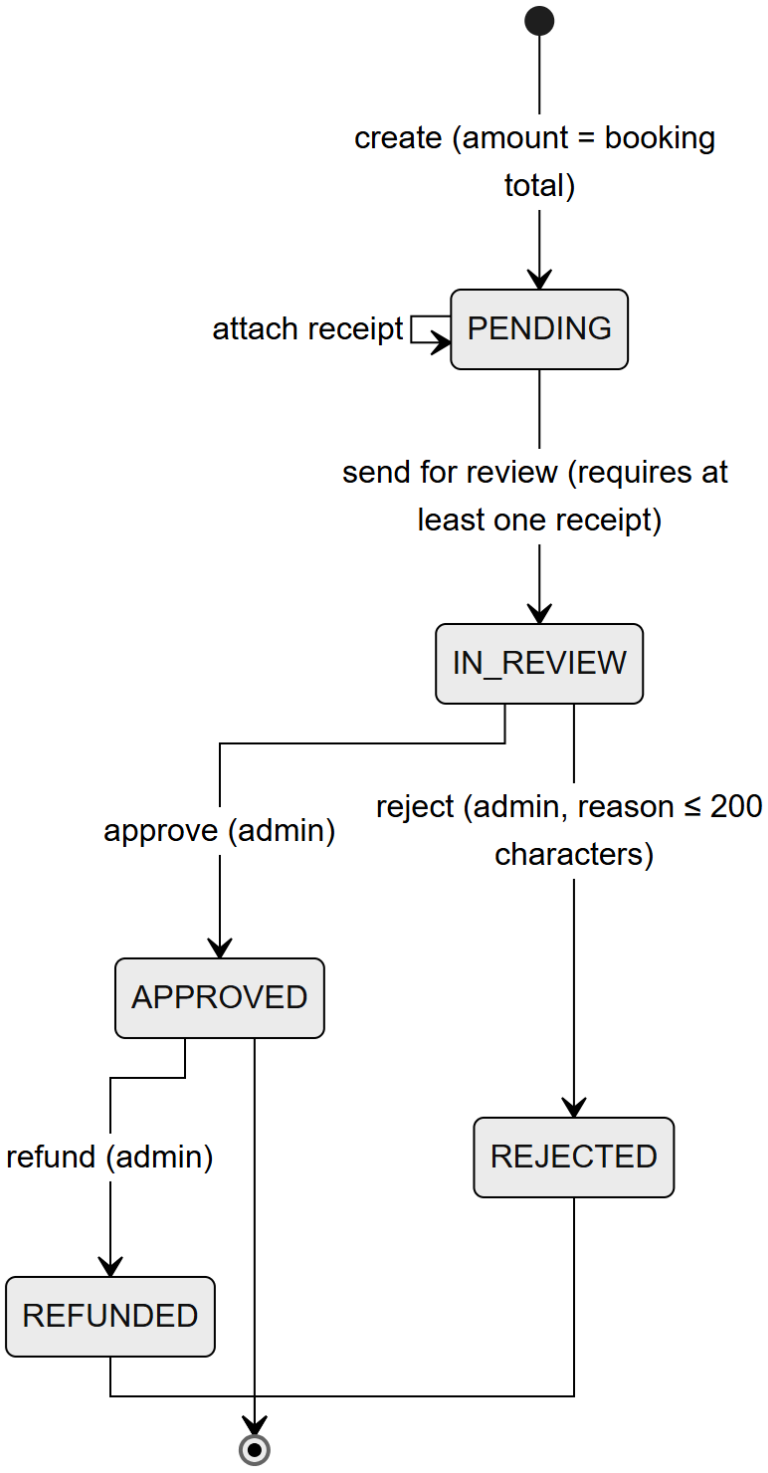


Note. The uppercase codes are the ones the system stores (ADR-010). Only `SCHEDULED` and `CONFIRMED` can be rescheduled or cancelled (RF-007). `SCHEDULED`, `CONFIRMED` and `IN_PROGRESS` take up the bay. `SCHEDULED` exists in the model, but the current flow never creates a booking in

that status: every booking starts directly as `CONFIRMED`. Source: `booking-service`, `Booking.moveTo`, `Booking.cancel`, `BookingStatus`.

8.3 Payment and Execution Life Cycles

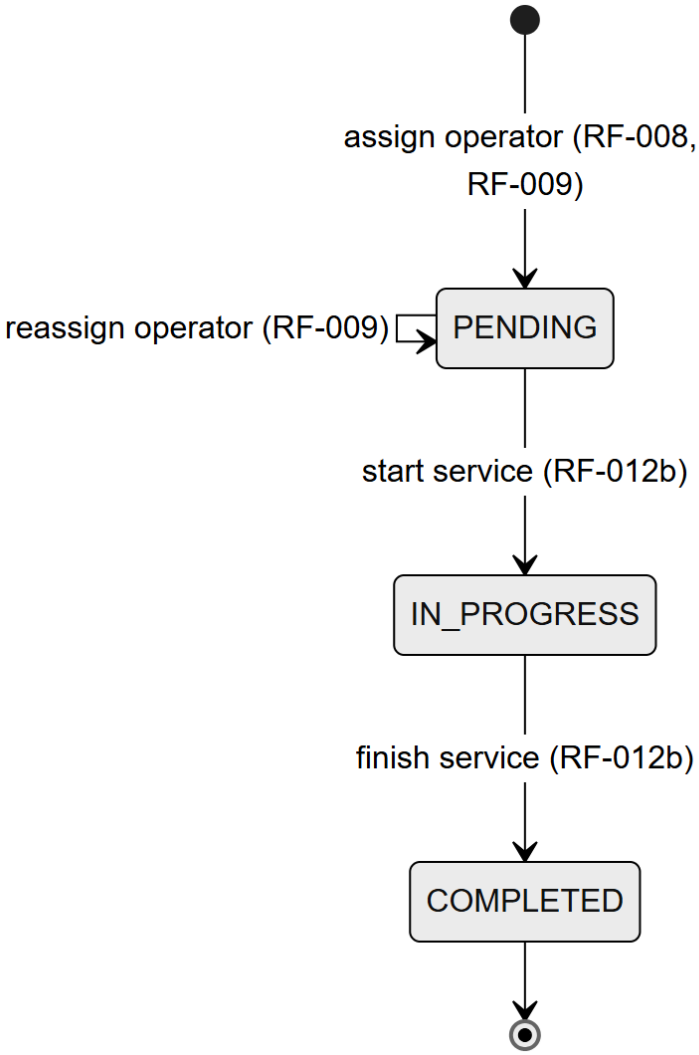
Figure 5 *Payment State Diagram*



Note. A payment reported by the customer moves from `PENDING` to `IN_REVIEW` in the same request; a manual (cash) payment goes straight to `APPROVED`. A booking allows only one approved payment (filtered unique index) and only one open payment (`PENDING`/`IN_REVIEW`) at a time. After

`REJECTED`, the customer can report a new payment for the same booking. Source: `payment-service`, `Payment` aggregate.

Figure 6 *Service Execution State Diagram*



Note. `COMPLETED` enables the customer's rating (RF-012). The status catalog also has `PAUSED` (Paused) and `WITH_ISSUE` (With issue), but no operation of the system moves an execution to those states. Source: `operations-service`, `ServiceExecution` (`reassign`, `start`, `finish`) and `ExecutionStatus`.

8.4 Domain Events and Notifications

The services publish events to RabbitMQ to notify other services without blocking the operation (ADR-004). Table 55 links each event to its effect.

Table 55 *Domain Events and Their Effect*

Event	Producer	Effect
`security.user_registered`	security-service	customer-service creates the profile (`customer.provisioned`); welcome notification.
`booking.confirmed`	booking-service	Published when a booking is created and when it is rescheduled. Booking notification and reminder scheduling; operations-service assigns the operator automatically, with the customer's preference if there is one.
`booking.cancelled`	booking-service	Cancellation notice and cancellation of the reminders.
`payment.confirmed`, `payment.rejected`	payment-service	Notification of the payment review result.
`payment.refunded`	payment-service	Refund notification to the customer.
`payment.promotion_redeemed`	payment-service	Notification of the redemption in the inbox and by email.
`payment.loyalty_points_earned`	payment-service	Notification of the points earned and, for each promotion those points unlock, a notice with its code.
`execution.operator_assigned`	operations-service	Notification to the customer and the operator; the booking no longer counts as unassigned.
`execution.service_started`, `execution.service_completed`	operations-service	Notification to the customer and cancellation of the pending reminders.
`execution.inspection_shared`	operations-service	Notification to the customer with the inspection report link.

Note. Each notification is written in the language of the account that receives it. The `security.user.deactivated`, `customer.provisioned` and `customer.vehicle.registered/updated/removed` events exist in the domain, but today they are only written to the service log: they are not published to RabbitMQ because no service consumes them. notification-service also listens to `booking.created`, which booking-service does not publish. The alert for bookings without an operator does not depend on an event: notification-service periodically checks the confirmed bookings that start in less than 2 hours without an operator.

9. Data Requirements

9.1 Data Ownership

The database has 9 schemas and 57 tables. Each schema has a single owner service, which is the only one allowed to write to it (see Table 56).

Table 56 *Data Schemas and Owner Service*

Schema	Owner service	Main data
`security`	security-service	People, addresses, accounts with their registration date, roles, permissions, sessions, lockouts after failed attempts, recovery codes and preferences.
`customer`	customer-service	Customers, vehicles and vehicle types.
`catalog`	booking-service	Categories, services, prices by vehicle type and points per service.
`booking`	booking-service	Site with its legal, tax and service data, opening hours, exceptions, bays, bookings, lines, cancellation reasons and opening-hours history.
`execution`	operations-service	Operators with their specialty and skills, shifts, absences, executions with their rating, and inspection reports with their phases, findings and photos.
`payment`	payment-service	Payment methods and accounts, payments, receipts and points ledger.
`promotion`	payment-service	Promotions, discount types and redemptions per booking.
`notification`	notification-service	Notifications, types, push devices, reminders, booking tracking for the alerts and processed events.
`audit`	security-service	Change audit log.

9.2 Integrity and Retention Rules

- References between schemas of different services are stored as identifiers without a foreign key and are validated through the API.
- Critical constraints are also enforced in the database: a filtered unique index for a single approved payment per booking, an index on the reference code, and range constraints for scores and percentages.
- The prices and points of each booking line are a copy of the value in force when the booking was made.
- Passwords and recovery codes are stored only as a hash.

- The public link of the inspection report uses a random code and does not show the customer's email, phone or full license plate.
- Records with history are not deleted; they are deactivated. This version does not define a retention or deletion policy for personal data, which must be defined before a real deployment in accordance with Law 1581 of 2012.
- Every schema change is applied with a versioned Liquibase migration.

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10. Traceability Matrix

Table 57 links each functional requirement to its user story, its use cases, the service that implements it and its status. The requirements without a user story (RF-027 to RF-038) describe built functions that must be added to the backlog.

Table 57 *Functional Requirements Traceability*

RF	Name	HU	CU	Service	Status
RF-001	Customer registration	HU-AUTH-001	CU02	security	Implemented
RF-002	Sign-in	HU-AUTH-002	CU01	security	Implemented
RF-003	Password recovery	HU-AUTH-003	CU03	security	Implemented
RF-004	Password change	HU-AUTH-004	CU04	security	Implemented
RF-004b	Lockout after failed attempts	HU-AUTH-002	CU01	security	Implemented
RF-005	Booking request	HU-BOOK-001	CU05	booking	Implemented
RF-006	Availability check	HU-BOOK-001	CU06	booking	Implemented
RF-007	Rescheduling and cancellation	HU-BOOK-002	CU07	booking	Implemented
RF-008	Automatic operator assignment	HU-BOOK-003	CU08	operations, booking	Implemented
RF-009	Manual assignment and reassignment	HU-BOOK-006	CU09	operations	Implemented
RF-010	Operator choice by the customer	HU-BOOK-004	CU05, CU08	operations, booking	Implemented

RF	Name	HU	CU	Service	Status
RF-011	Assigned services	HU-BOOK-007	CU10	operations	Implemented
RF-012	Service rating	HU-BOOK-005	CU15	operations	Implemented
RF-012b	Service start and finish	HU-BOOK-007	CU11	operations	Implemented
RF-013	Payment methods	HU-PAY-001	CU12	payment	Implemented
RF-013b	Receipt upload	HU-PAY-001	CU12	payment	Implemented
RF-014	Payment review	HU-PAY-003	CU13	payment	Implemented
RF-015	Payment history	HU-PAY-002	CU14	payment	Implemented
RF-016	Payment constraints	HU-PAY-001	CU12	payment	Implemented
RF-017	Dashboard per role	HU-UI-001	CU16	frontend	Implemented
RF-018	Notifications	HU-UI-002	CU17, CU32	notification	Implemented
RF-019	Visual theme per account	HU-UI-003	CU18	security	Implemented
RF-020	Language per account	HU-UI-003	CU19	security	Implemented
RF-021	Opening hours and exceptions	HU-ADMIN-001	CU20	booking	Implemented
RF-022	Operator availability	HU-ADMIN-001	CU22	operations	Implemented
RF-023	Operational reports	HU-ADMIN-001	CU23	booking, operations, payment	Partial
RF-024	Promotion management	HU-PROMO-001	CU24	payment	Implemented
RF-025	Applying a promotion at payment	HU-PROMO-002	CU25	payment	Implemented
RF-026	Loyalty points	HU-PROMO-002	CU25	payment	Implemented
RF-027	Own account management	—	CU31	security	Implemented
RF-028	User administration	—	CU27	security	Implemented
RF-029	Roles and permissions	—	CU27	security	Implemented
RF-030	Customer vehicle management	—	CU26	customer	Implemented
RF-031	Service catalog and prices	—	CU21	booking	Implemented
RF-032	Bay management	—	CU20	booking	Implemented
RF-033	Administrative booking management	—	CU29	booking	Implemented
RF-034	Site payment accounts	—	CU28	payment	Implemented
RF-035	Refund	—	CU13	payment	Implemented
RF-036	Vehicle inspection report	—	CU30	operations, notification	Implemented
RF-037	Business data	—	CU20	booking	Implemented

RF	Name	HU	CU	Service	Status
RF-038	Operator profile	—	CU22	operations	Implemented

11. Progress Status and Open Decisions

11.1 Progress Summary

Of the 41 functional requirements, 40 are implemented and 1 is partially implemented (RF-023, consolidated reports). The seven server components are built; on October 9, 2026 the six services and the gateway were started, and each one answered `200 OK` on its health endpoint. The web and the Android app are connected to the six services through the gateway. Table 58 summarizes the automated tests of the services and the web, run on October 19, 2026.

Table 58 *Automated Tests by Component*

Component	Command	Tests	Count date
Web application (Angular)	`npx ng test --watch=false`	43	October 19, 2026
api-gateway	`./mvnw test`	34	October 19, 2026
security-service	`./mvnw test`	151	October 19, 2026
customer-service	`./mvnw test`	69	October 19, 2026
booking-service	`./mvnw test`	43	October 19, 2026
operations-service	`./mvnw test`	55	October 19, 2026
notification-service	`./mvnw test`	112	October 19, 2026
payment-service	`dotnet test`	148	October 19, 2026
Total		**655**	

Note. All tests passed. The 148 of payment-service include 7 persistence tests that need SQL Server; on October 19 they were run against an isolated database. In addition, `lavarapido-infra/e2e` has 19 end-to-end tests that run with the system up and passed on October 19, 2026, and the 42 functional cases of the Test Documentation passed that same day: the booking flow with operator choice and automatic assignment, the complete flow of the three roles (booking, payment, approval with points, service, and rating), the inspection report with photos and public link, and the access rules per role. The folder also includes the RNF-001 measurement and the check of the web screens at 360 px (RNF-008). During the manual review of October 9, a

missing API Gateway route (`/api/v1/loyalty/**`) and a style that misaligned the promotion badges were fixed.

11.2 Pending Items for Delivery

- Complete the consolidated reports for a chosen period (RF-023).
- Write the user stories for RF-027 to RF-038.
- Write integration tests against SQL Server for the concurrency rules (RNF-003).
- Test the app on an Android 8 device (RNF-008) and run a usability test with users (RNF-009).
- Add the end-to-end tests to the continuous integration; today they are run by hand with the system up.

11.3 Open Decisions

Table 59 *Open Decisions*

Topic	Current situation	Owner
Payment before the service	It has not been defined whether payment must be approved before the service starts.	Team and car wash manager
Permissions by role	The permission matrix is editable but is not applied to authorization (ADR-015).	Backend lead
Displayed promotion price	The price shown on the promotion card and its actual discount percentage are not tied together by any rule.	Team
Data retention	There is no retention or deletion policy for personal data.	Project lead

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